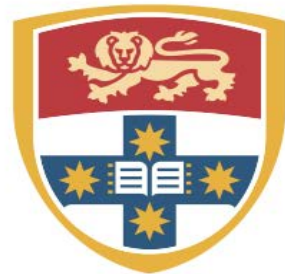


Phylogenetics and Molecular Dating in the Genomic Era

Simon Ho

School of Life and Environmental Sciences



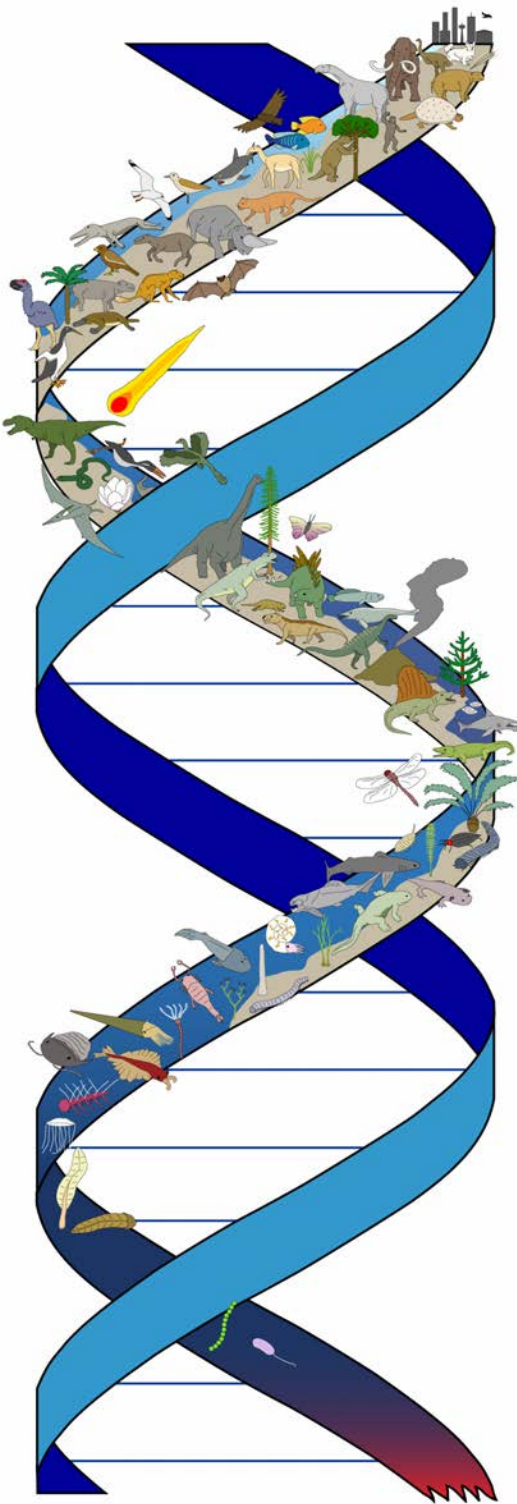
THE UNIVERSITY OF
SYDNEY





Molecular Ecology, Evolution, and Phylogenomics

- Phylogenomics
- Phylogenetic analysis
- Molecular evolution
- Molecular ecology
- Population genomics
- Molecular systematics



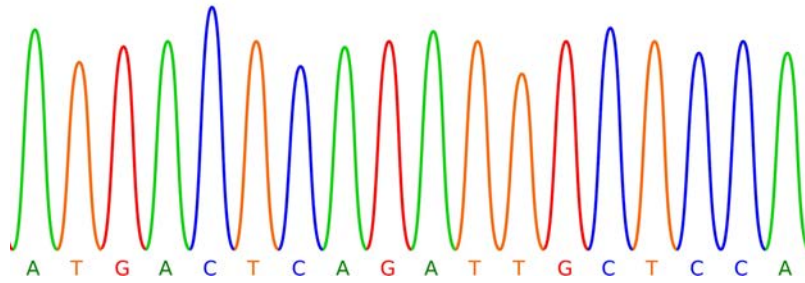
Outline

1. Phylogenomics
2. Concatenation or coalescence?
3. Strategies for large data sets
4. Phylogenomic dating
5. Example: Evolution of birds

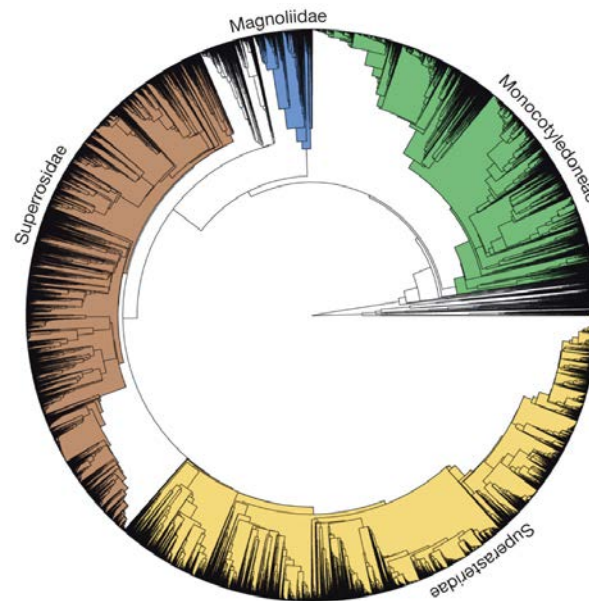
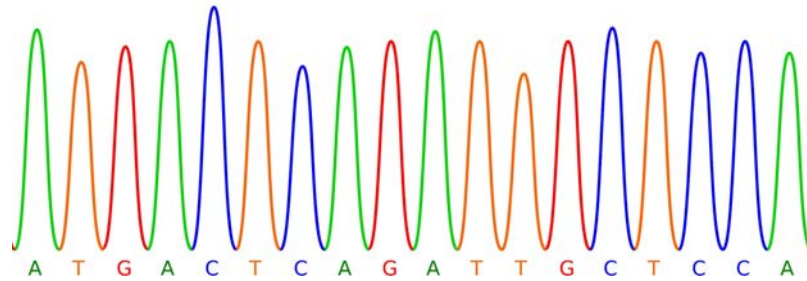
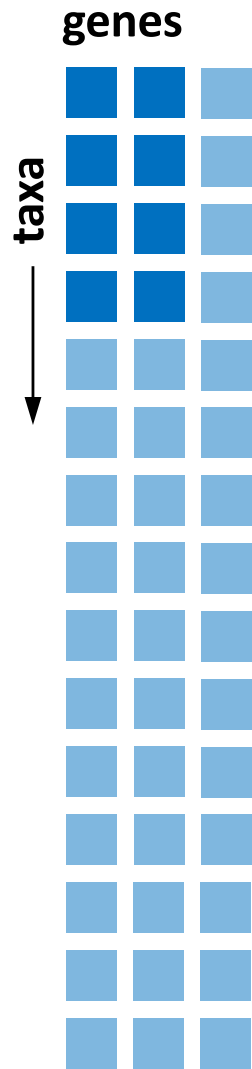
1. Phylogenomics

What is phylogenomics and what are the main challenges?

Molecular data sets



Molecular data sets



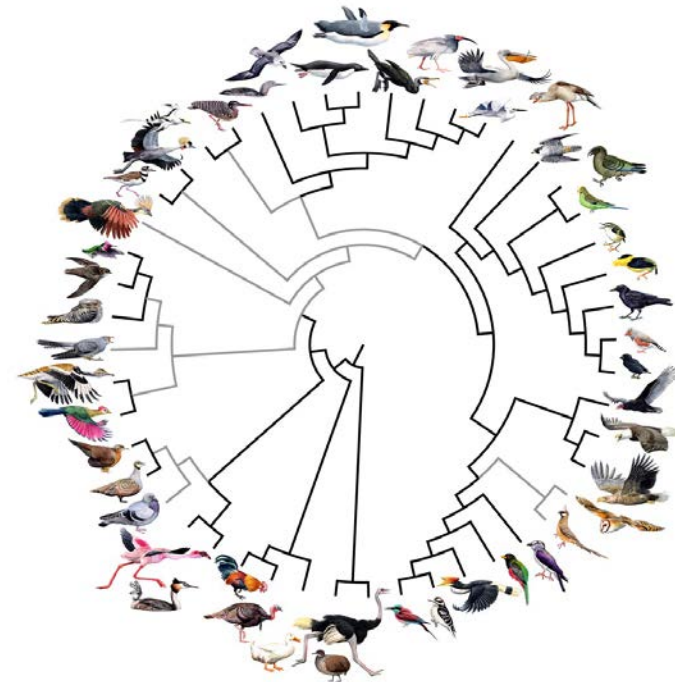
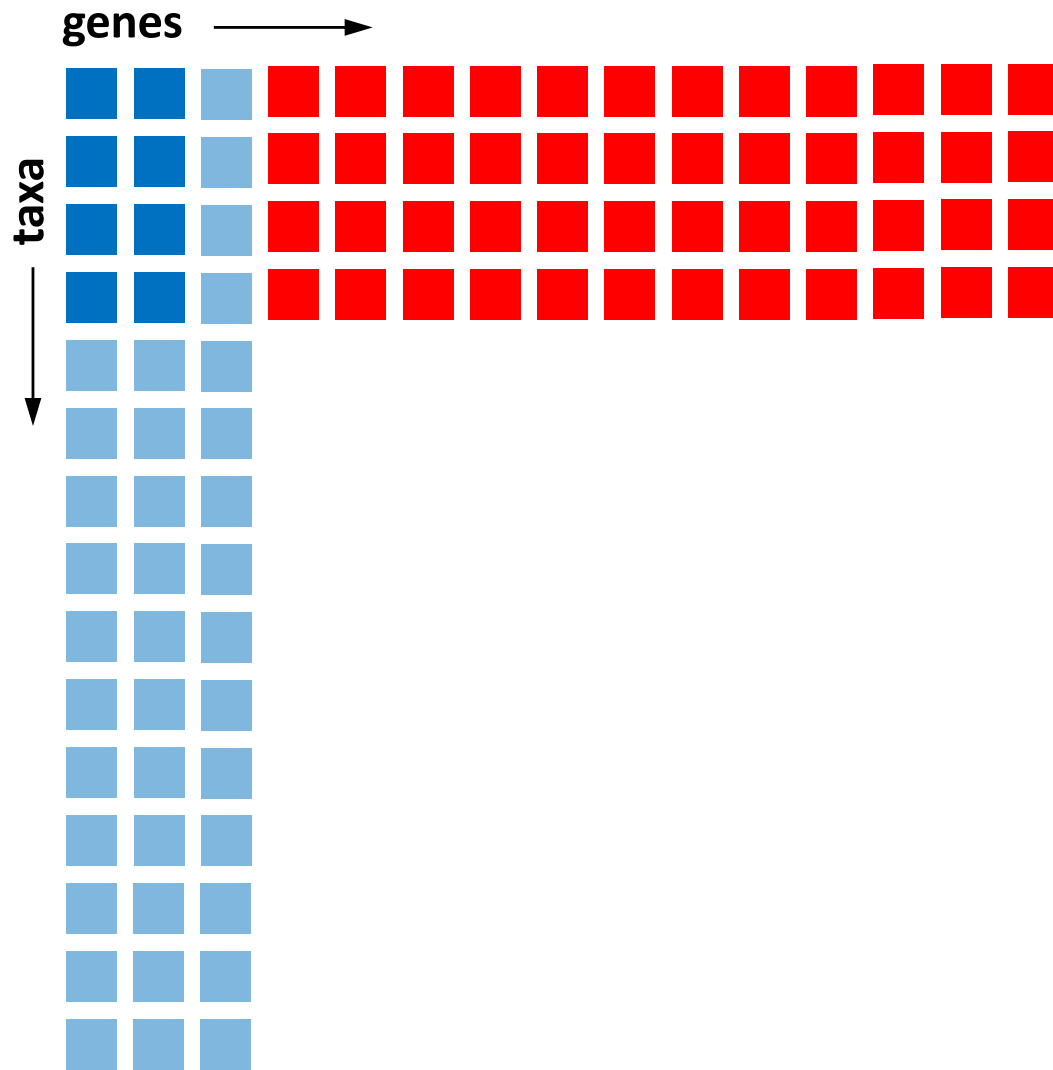
‘megaphylogenies’

Flowering plants

32,223 taxa

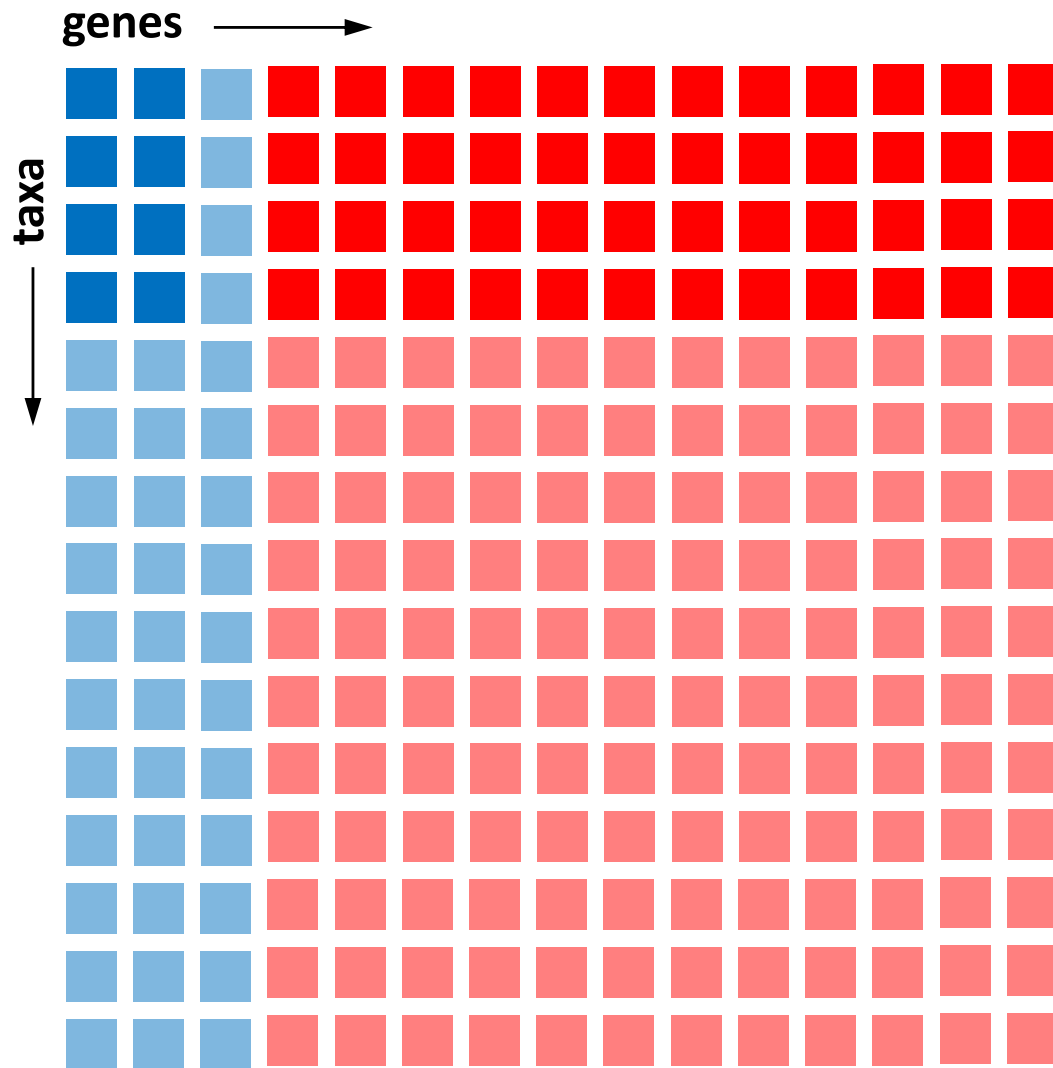
7 genes

Molecular data sets



Birds
48 taxa
8295 genes

Molecular data sets

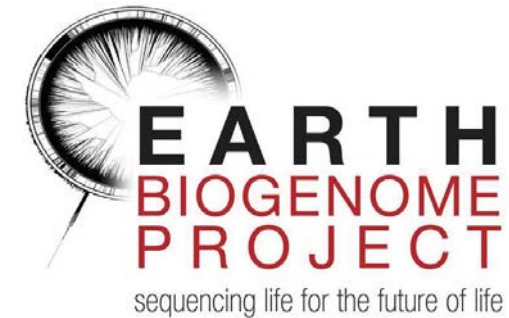
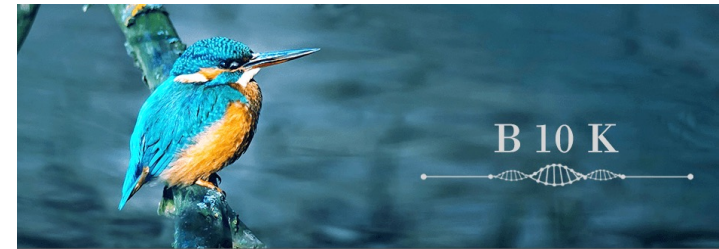
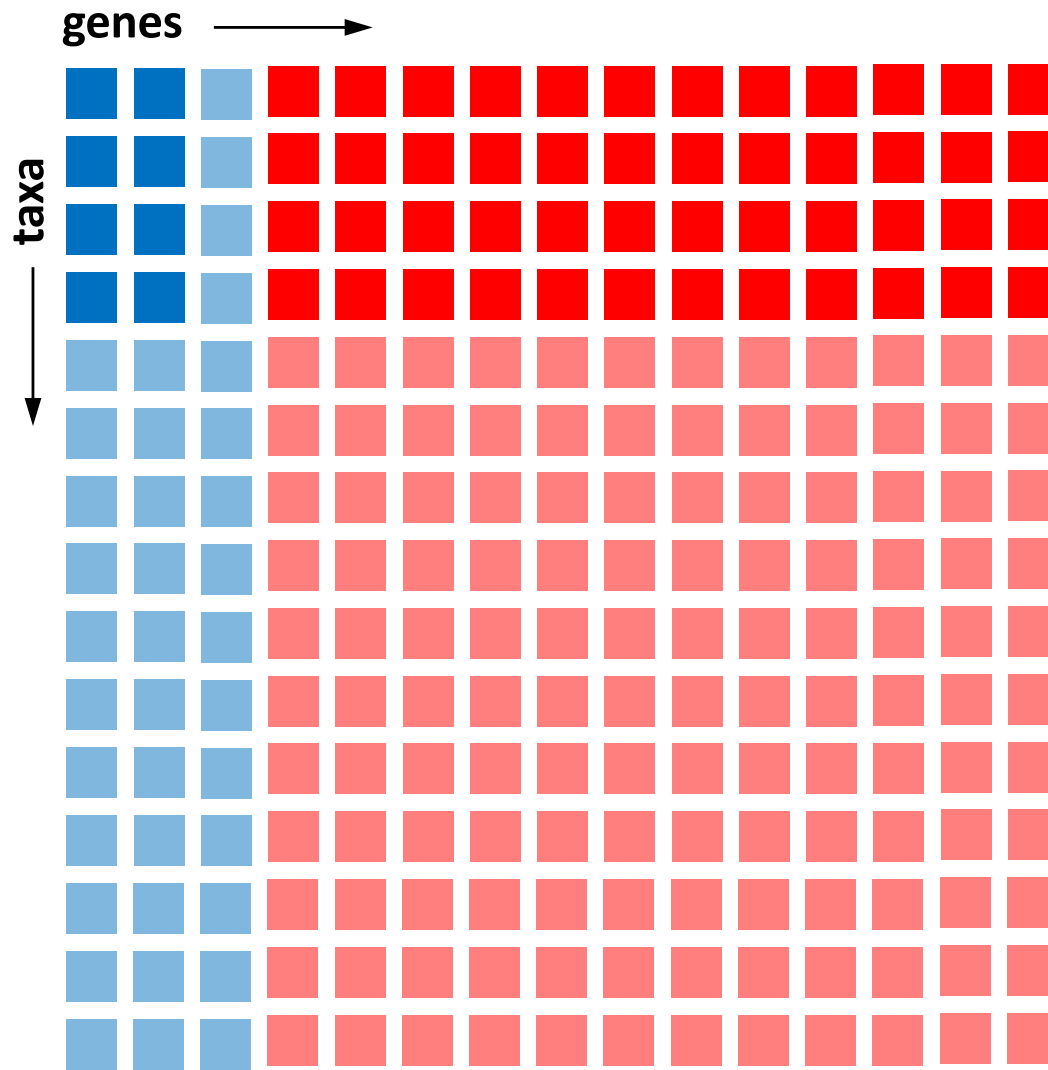


Flowering plants

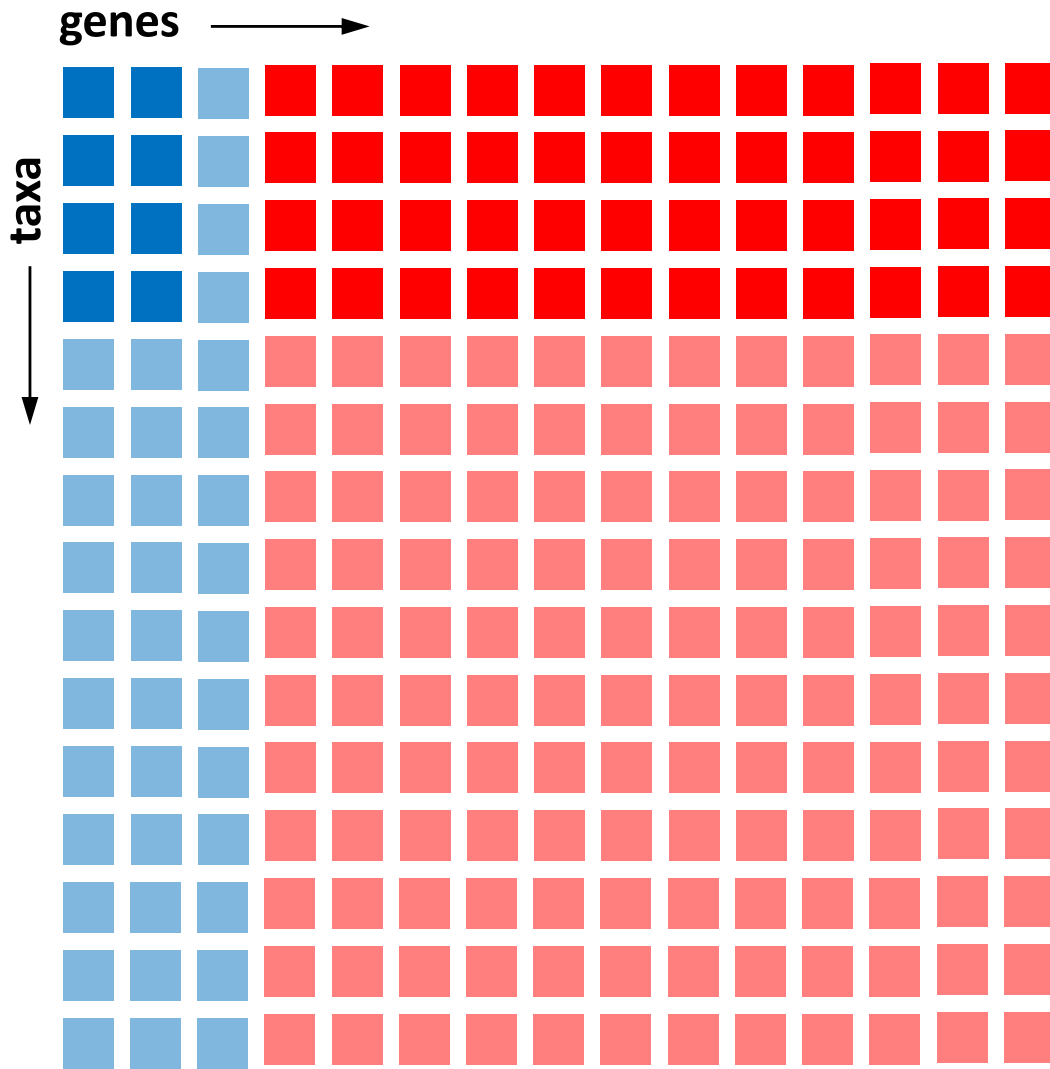
9506 species

353 genes

Molecular data sets



Molecular data sets

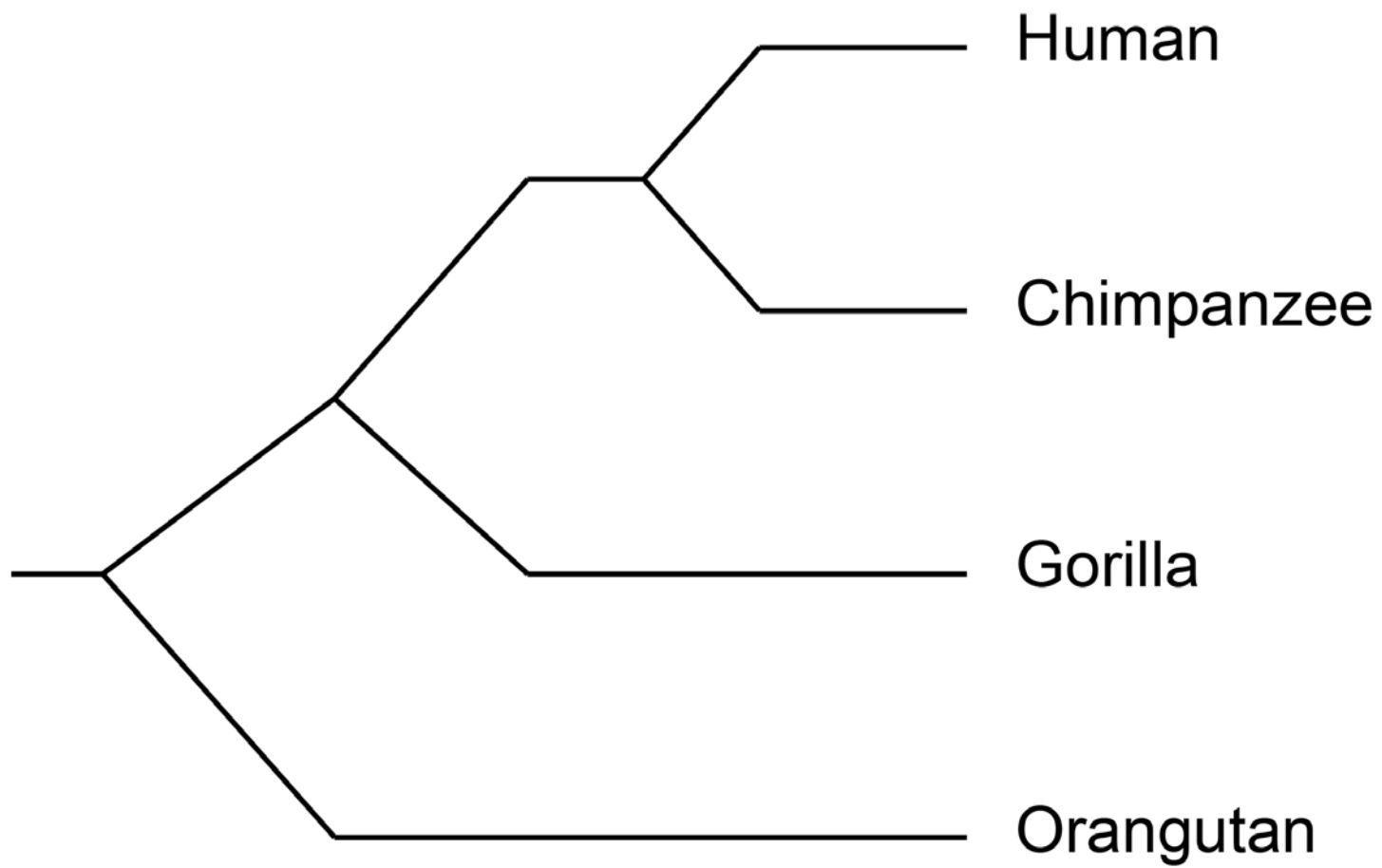


**Valuable source of
biological information**

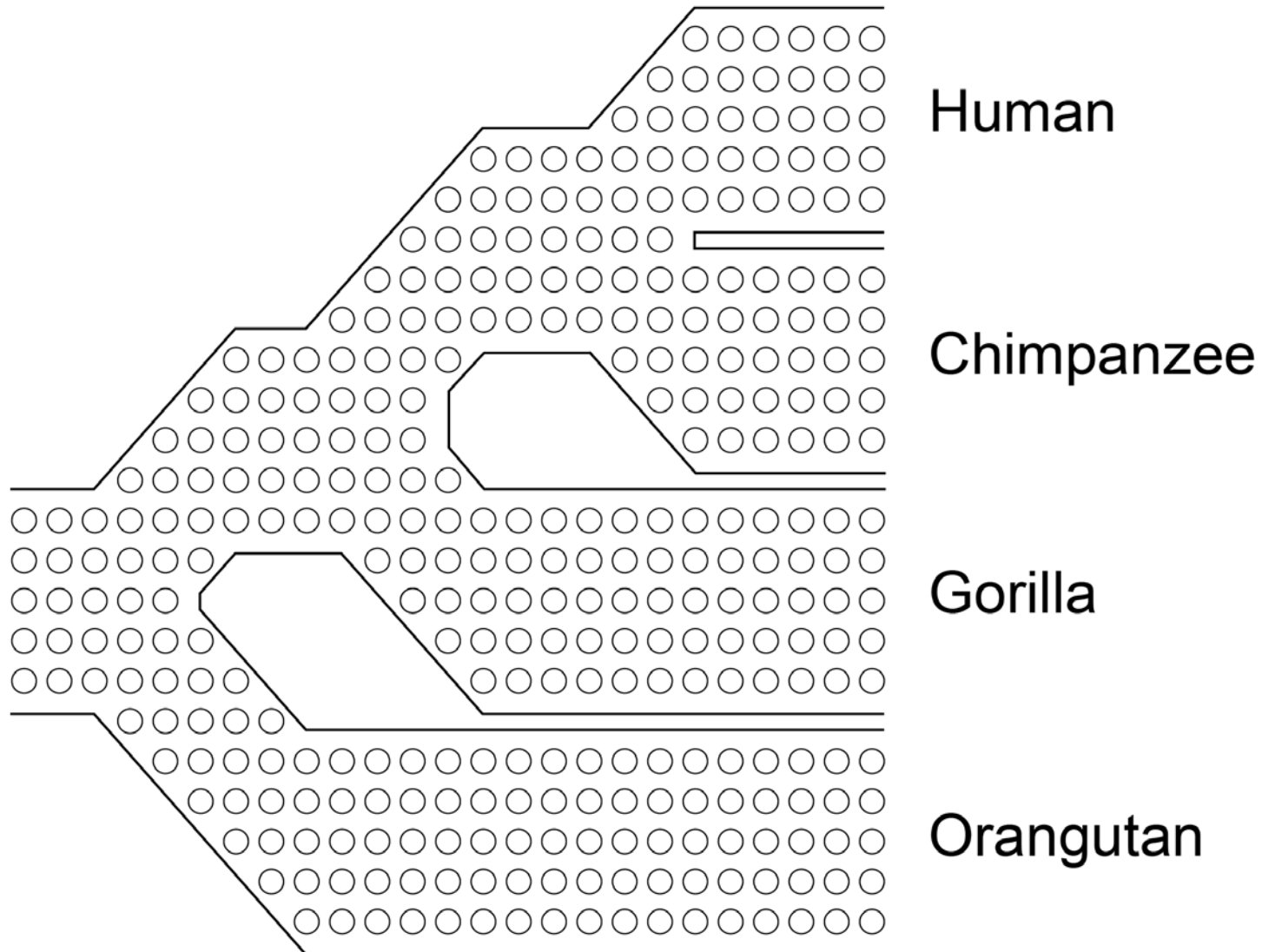
Challenges

- Analysis is computationally expensive
- Complex evolutionary patterns
- Evolutionary rate variation
- Discordant 'gene trees'

Species tree

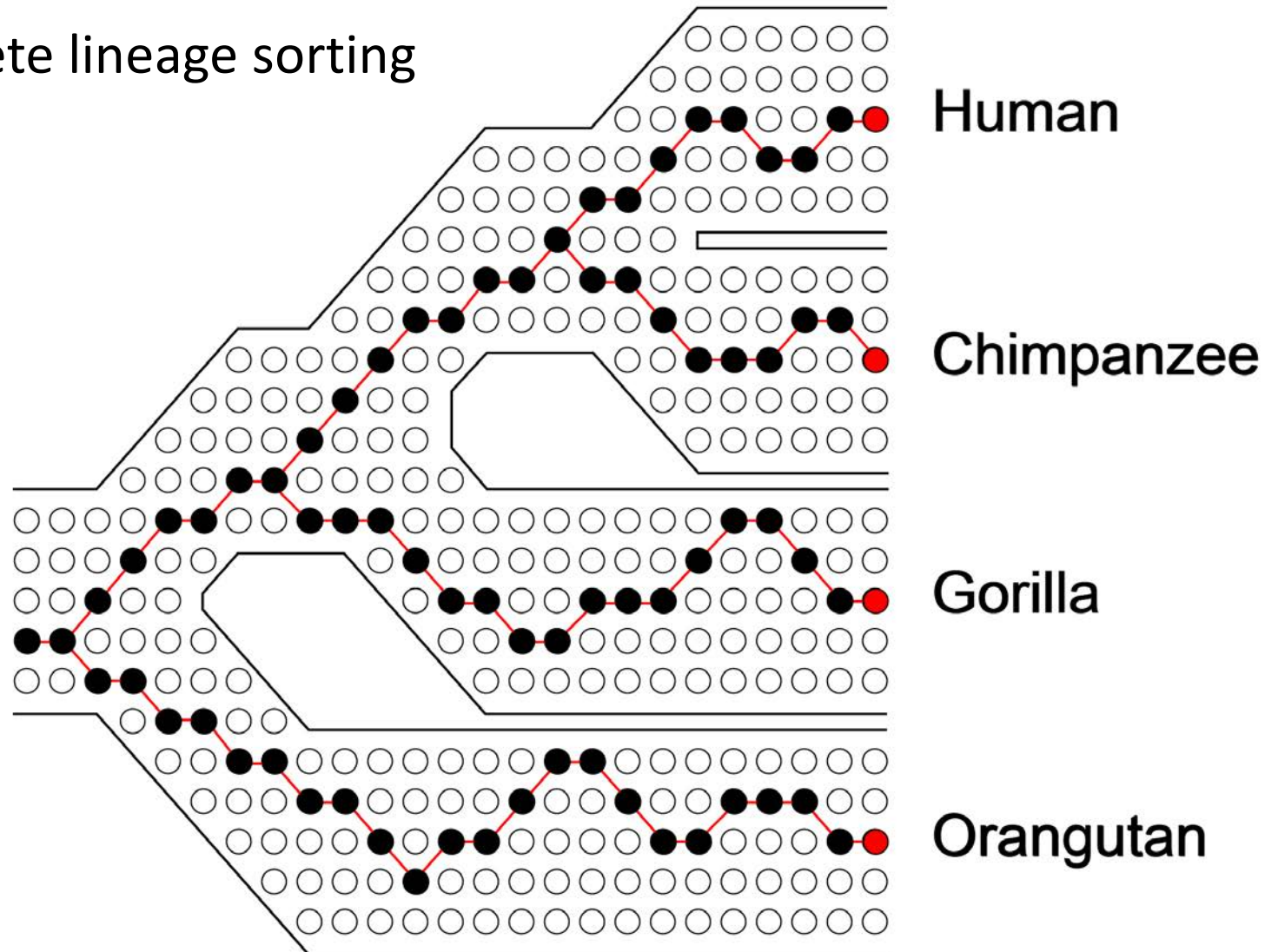


Multispecies coalescent



Gene tree (concordant)

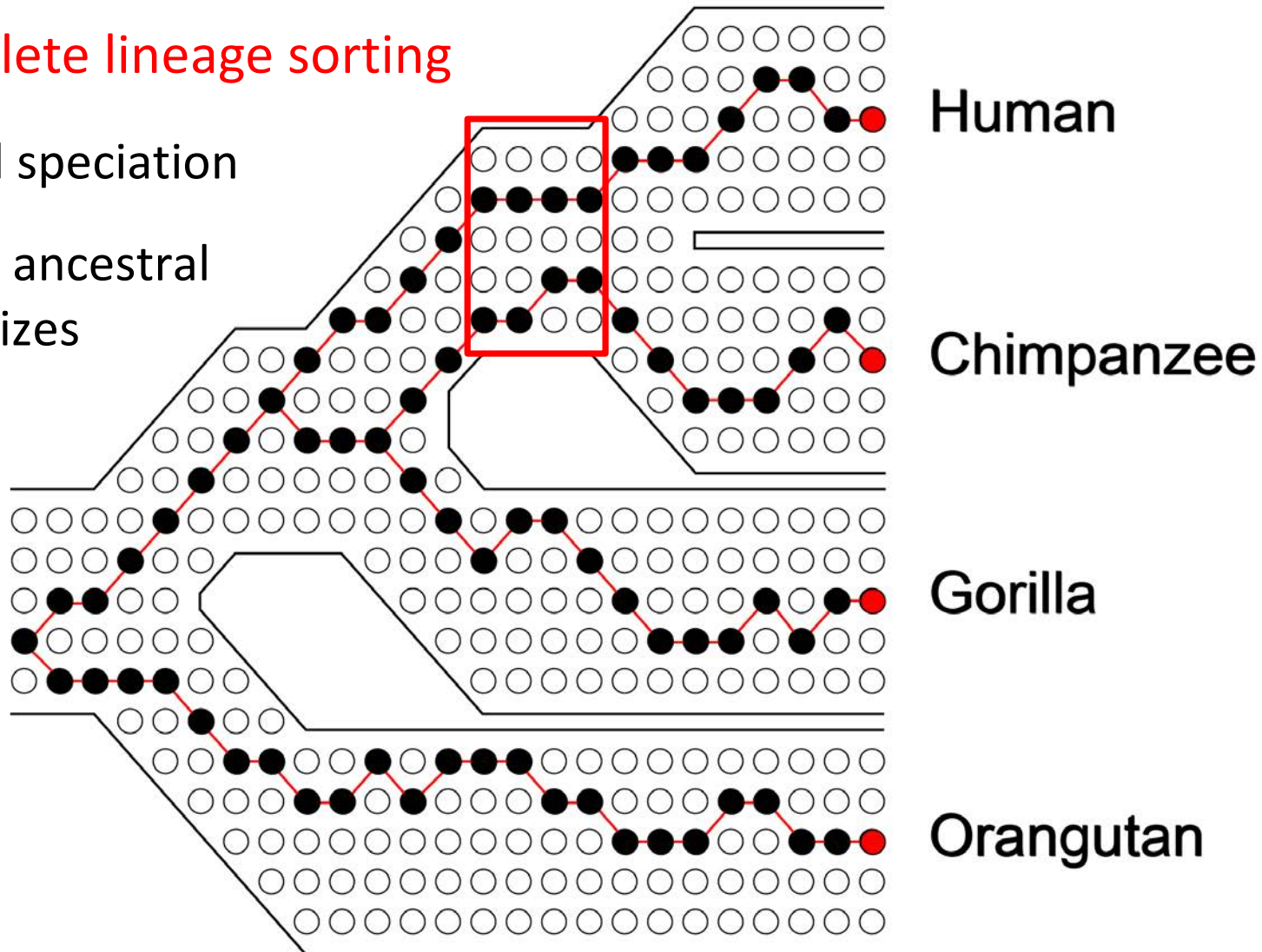
Complete lineage sorting



Gene tree (discordant)

Incomplete lineage sorting

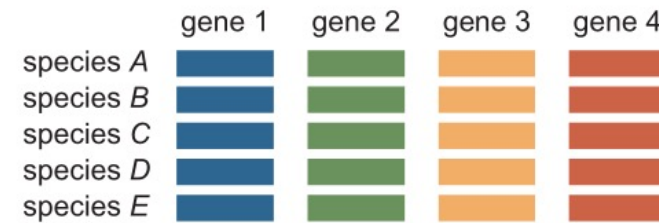
- Rapid speciation
- Large ancestral pop sizes



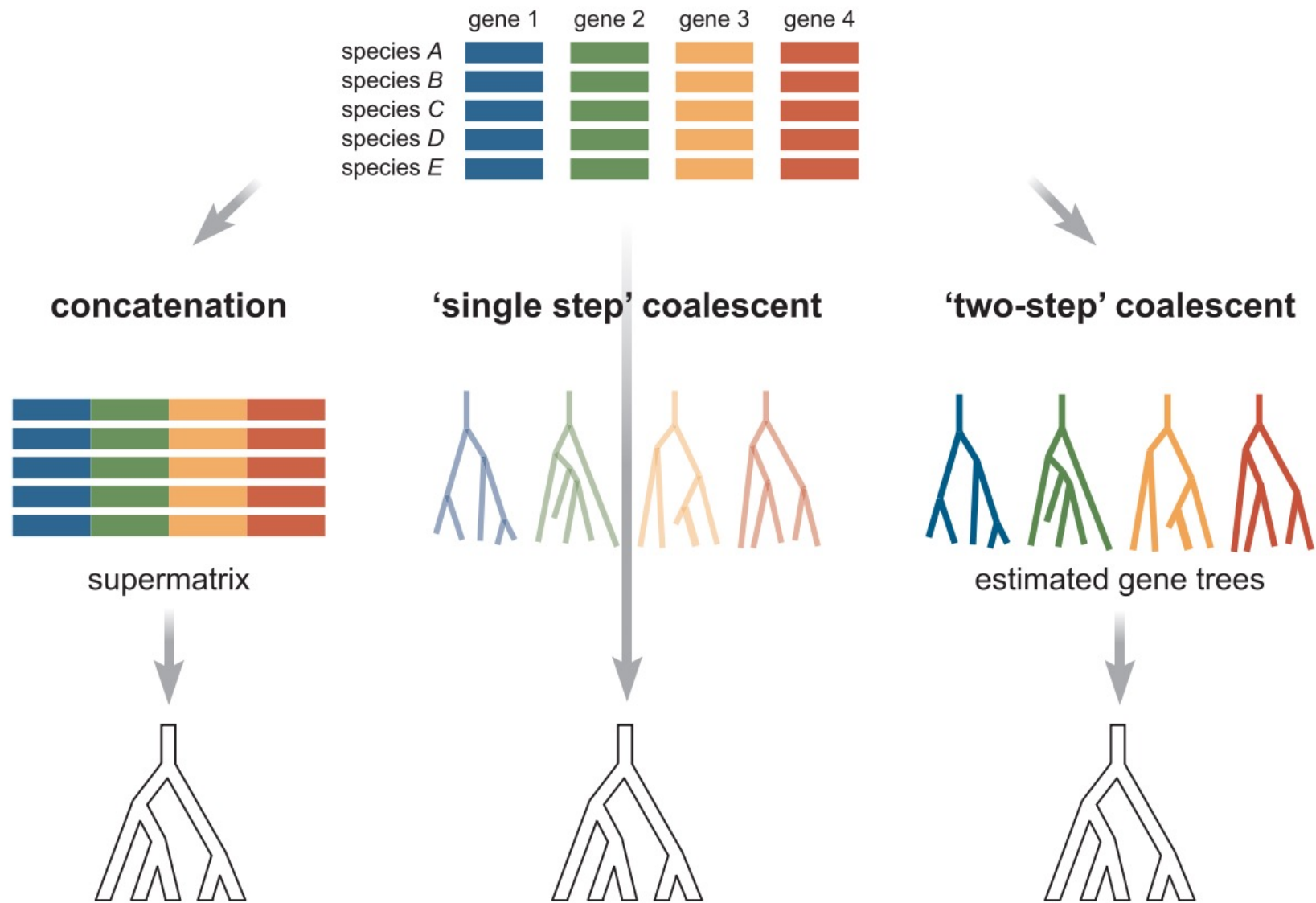
2. Concatenation or coalescence?

What is the best approach to analysing
phylogenomic data sets?

Phylogenomic analysis



Phylogenomic analysis

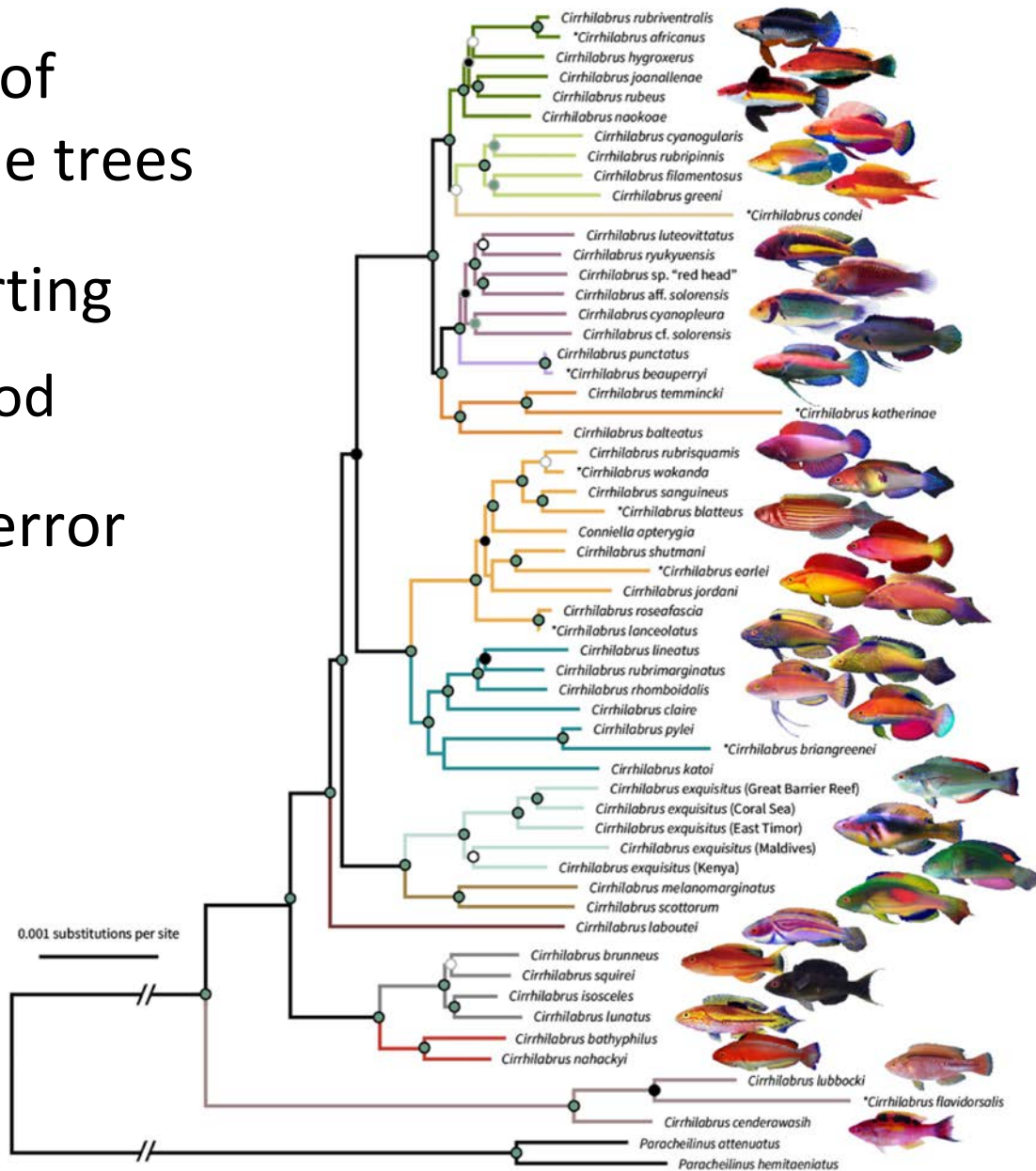


Concatenation or coalescence?

Consider the main cause of incongruence among gene trees

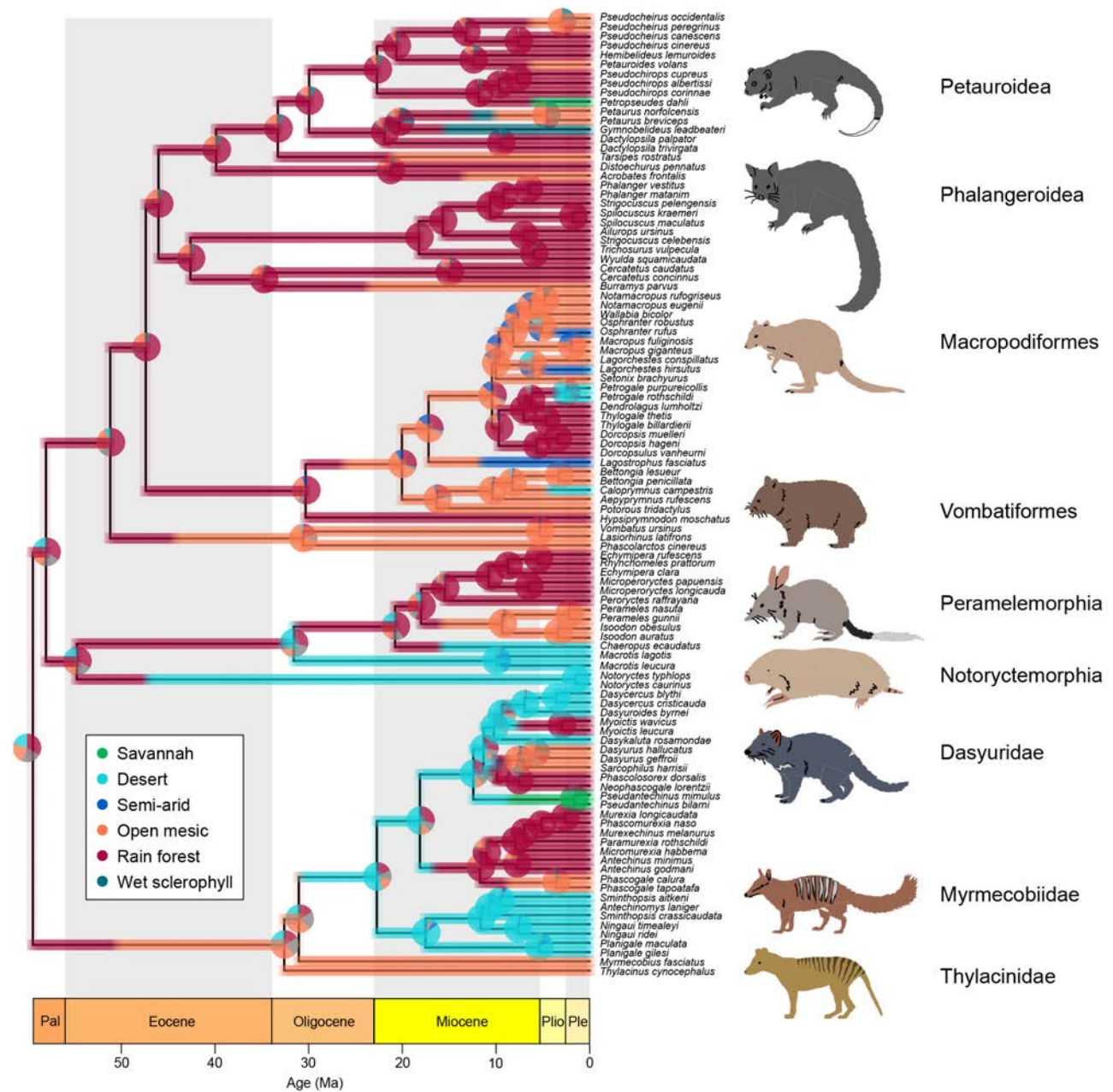
- Incomplete lineage sorting
 - Use coalescent method
- Gene-tree estimation error
 - Use concatenation

Example:
Fairy wrasses
(Labridae)



Concatenation or coalescence?

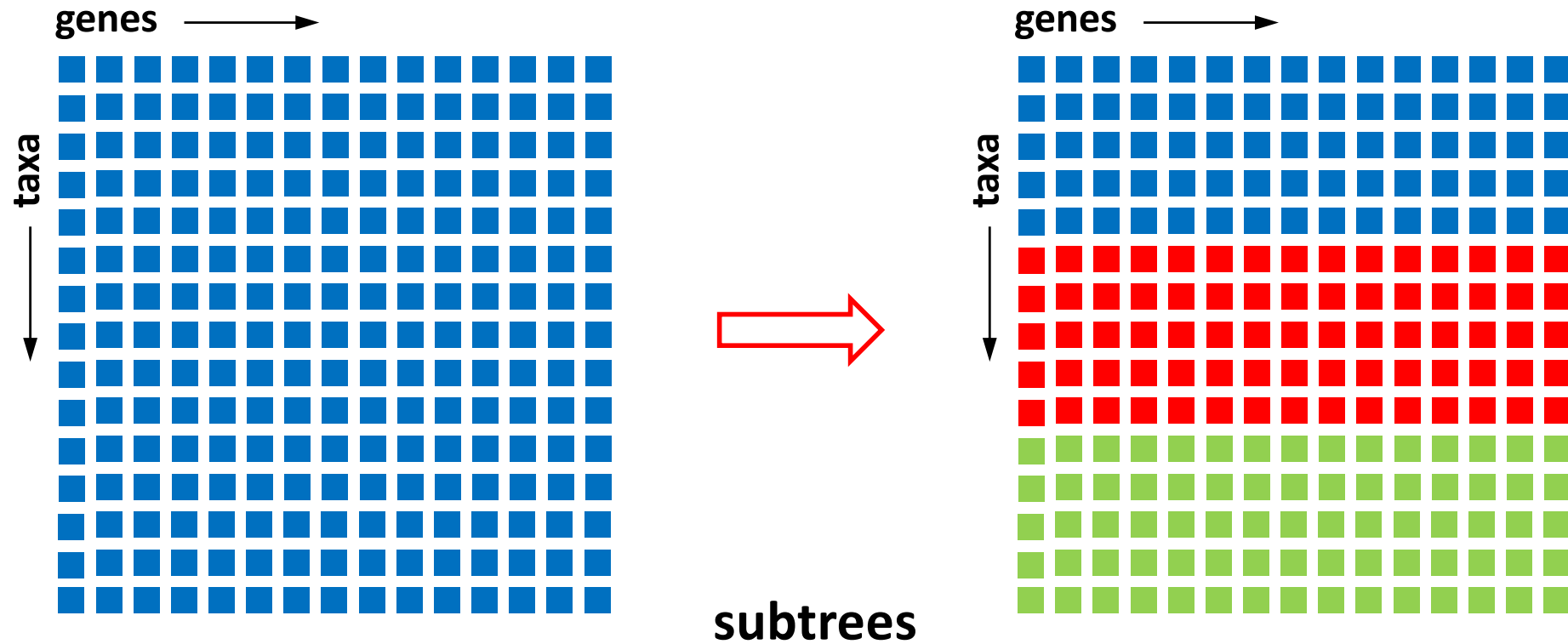
- 2039 exons
- 108 species (70 genera)



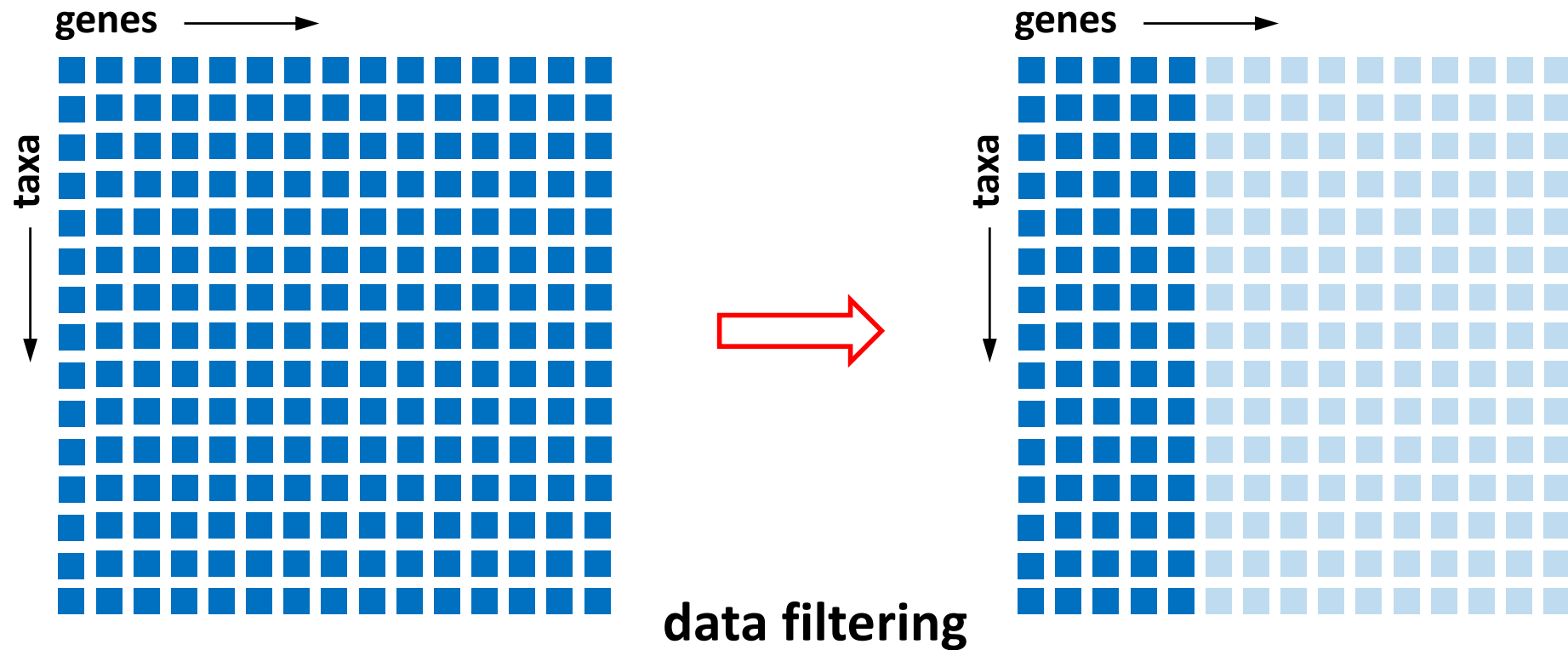
3. Strategies for large data sets

How can we handle large phylogenomic data sets?

Achieving computational feasibility



Achieving computational feasibility



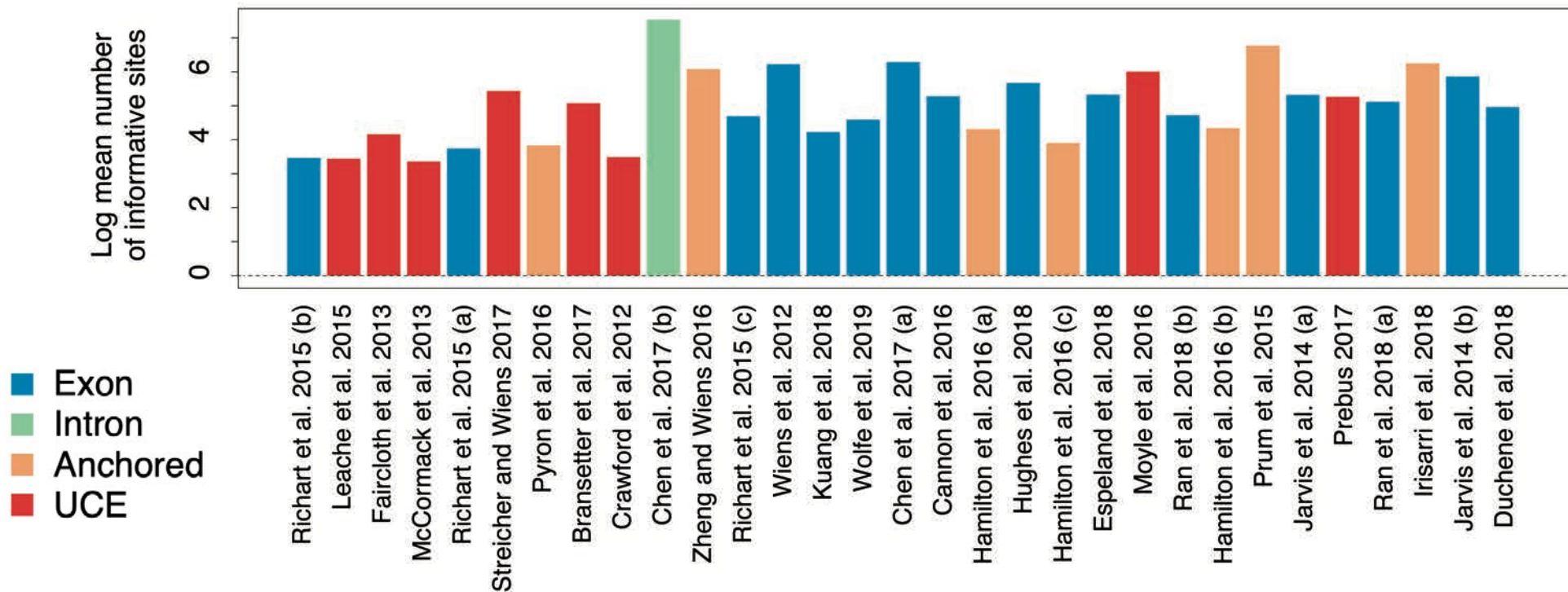


PhyloMA d

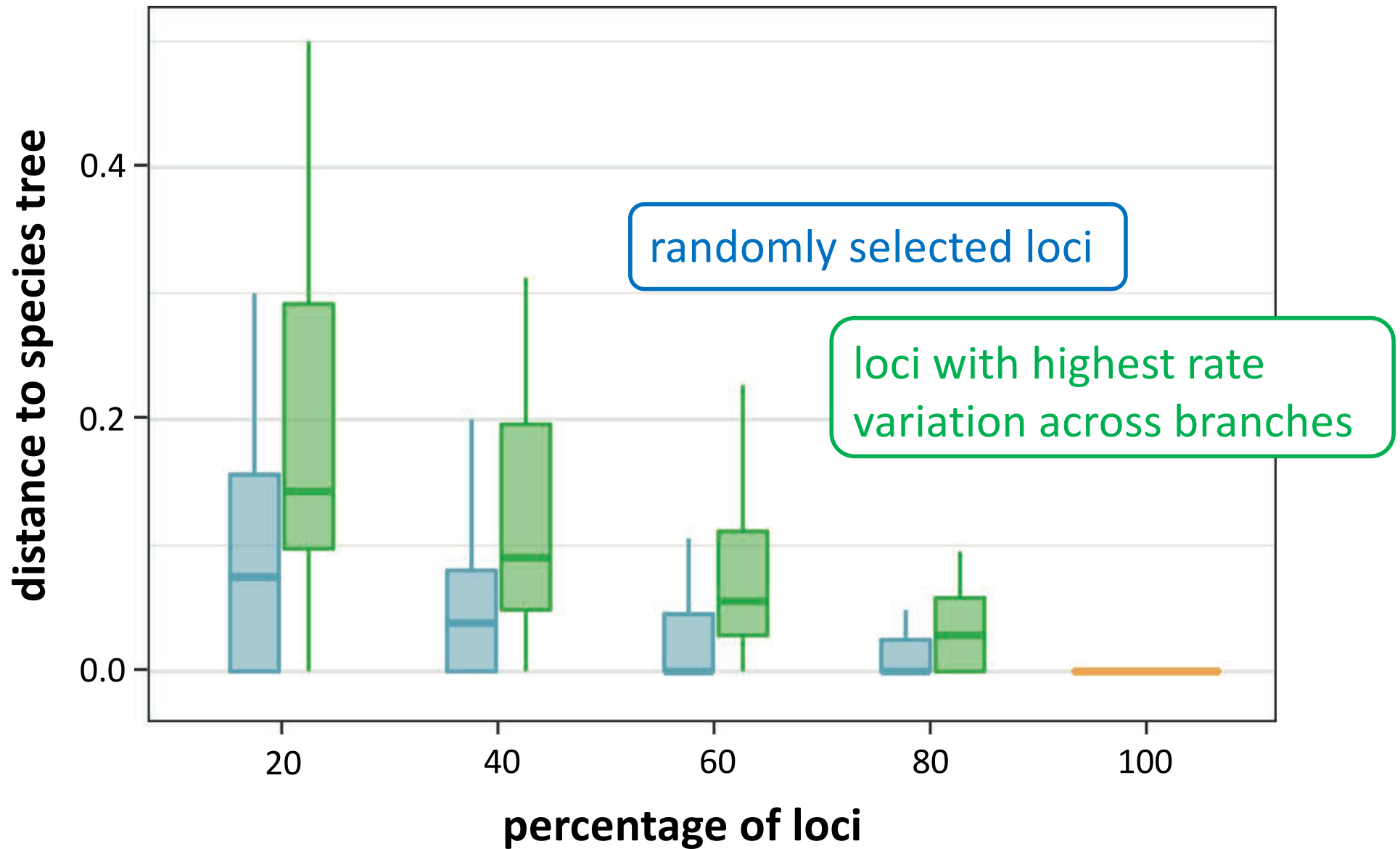
Efficient assessment of phylogenomic model adequacy

- Substitution saturation
- Compositional heterogeneity
- Substitution model adequacy
- Clock model adequacy

Substitution saturation



Evolutionary rate variation



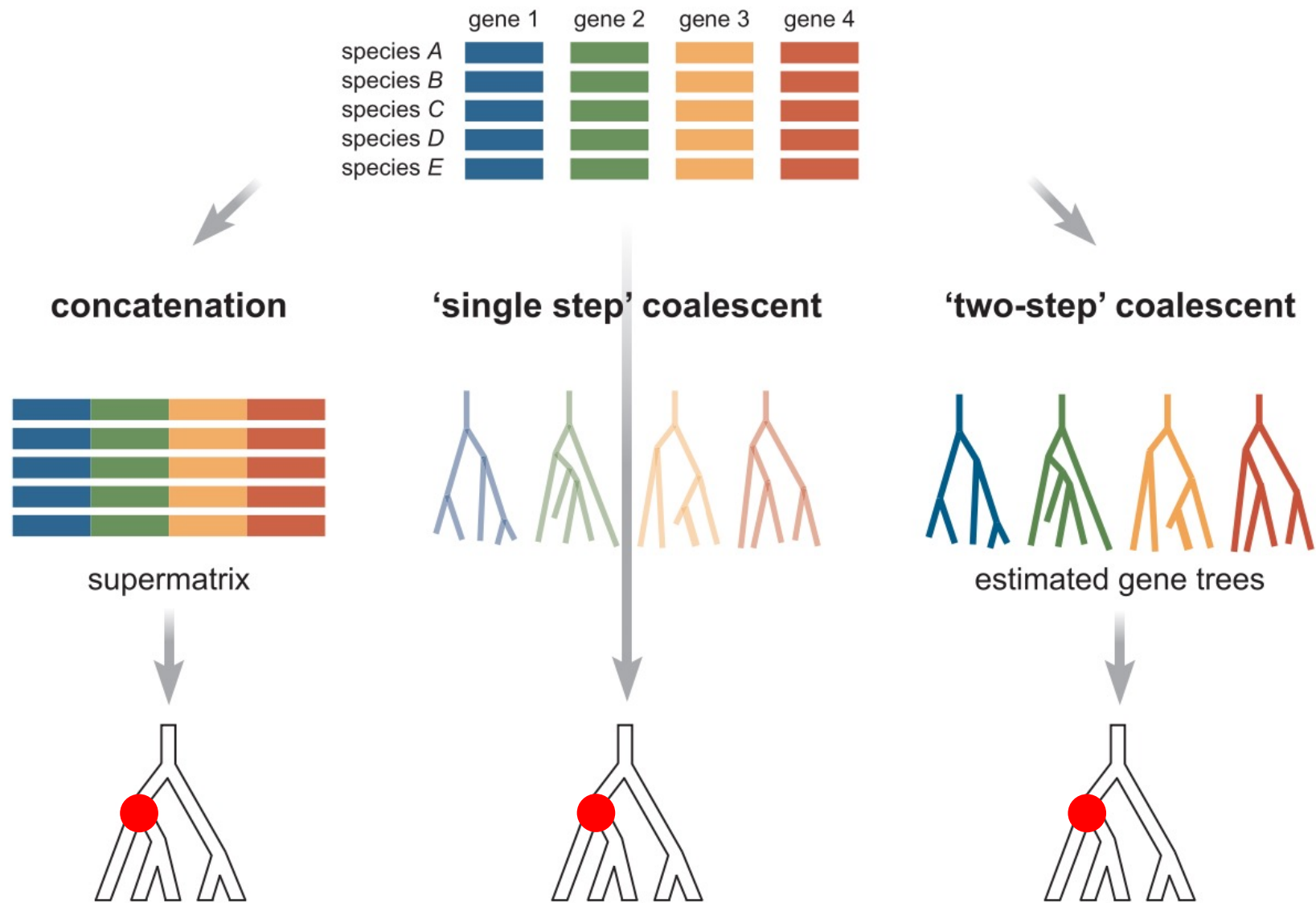
Phylogenomic data filtering

- **Rate variation across branches**
- **Substitution saturation**
- Compositional heterogeneity
- Phylogenetic signal
- Missing data
- Random

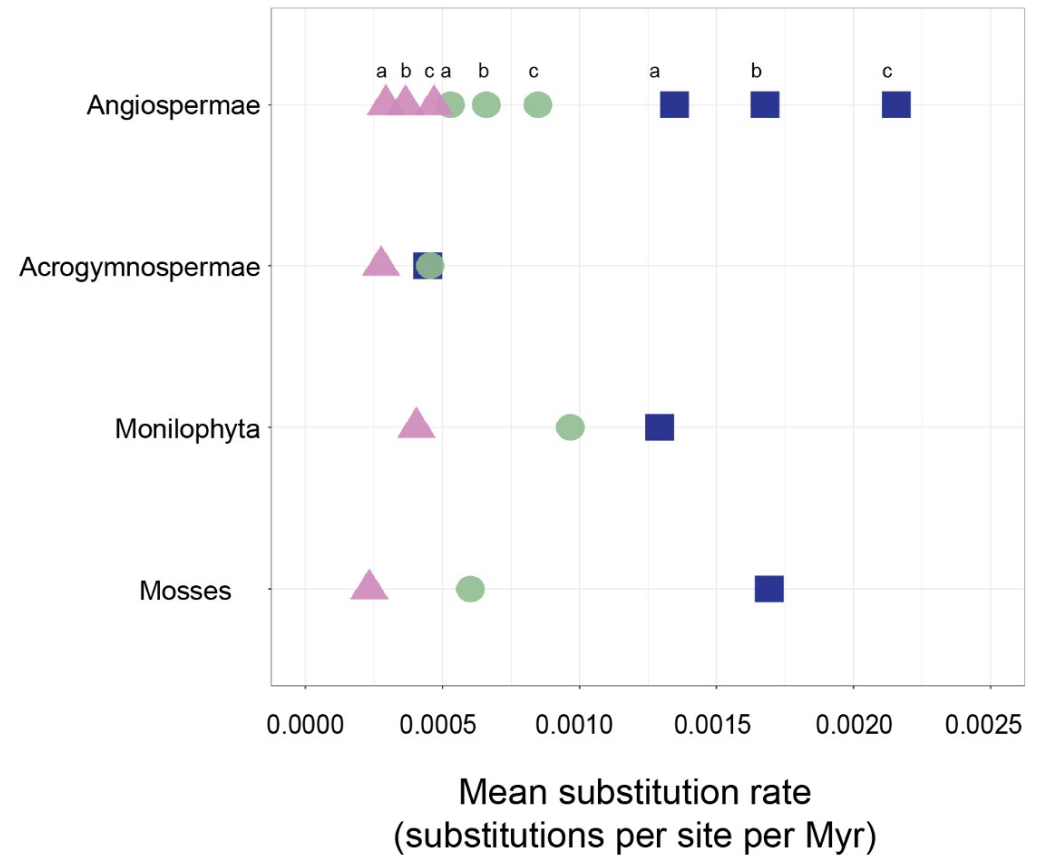
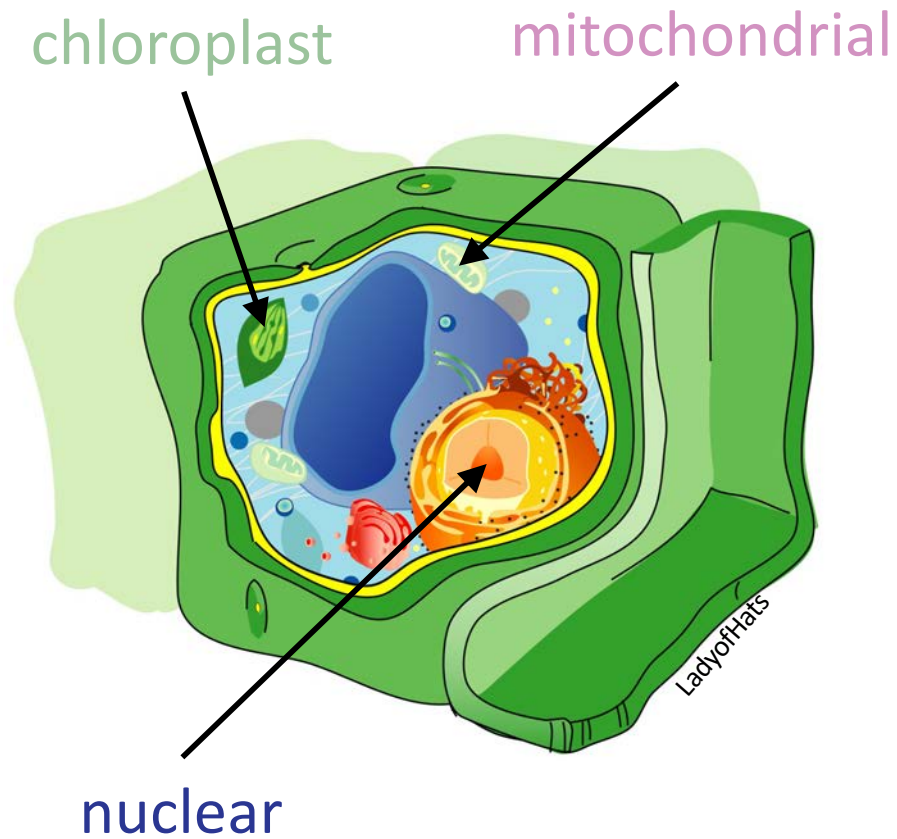
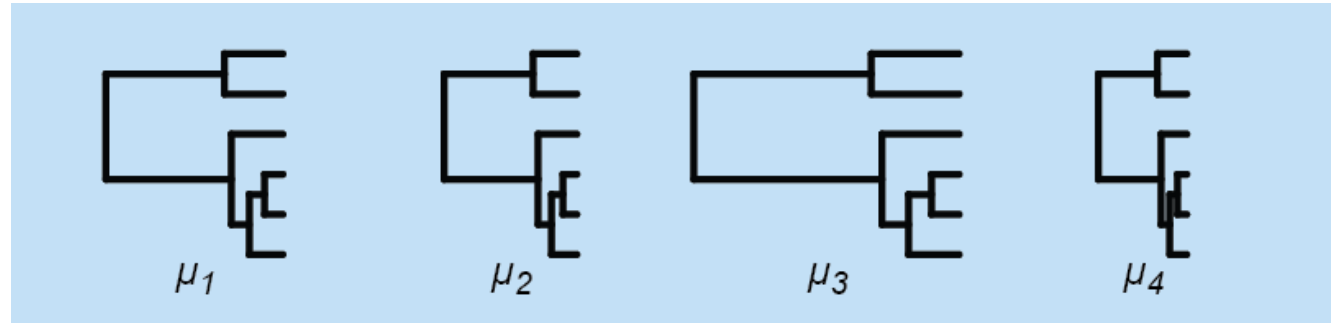
4. Phylogenomic dating

Strategies for molecular dating analyses
of phylogenomic data sets

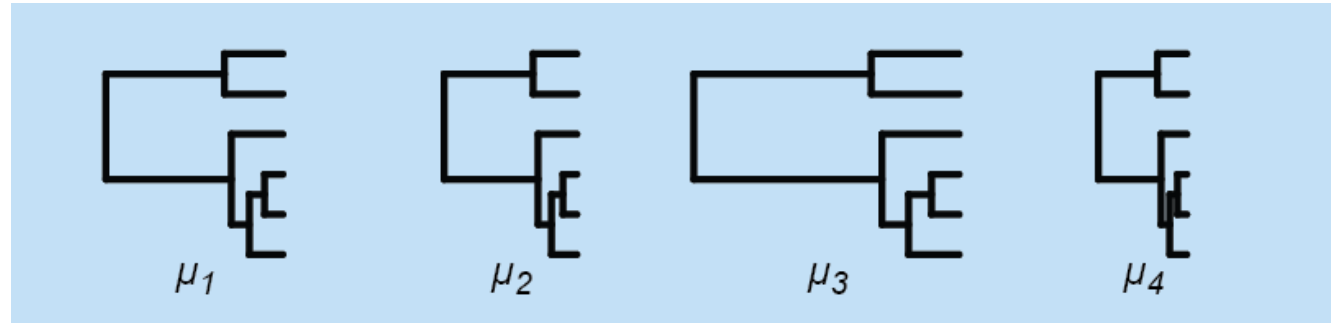
Phylogenomic dating



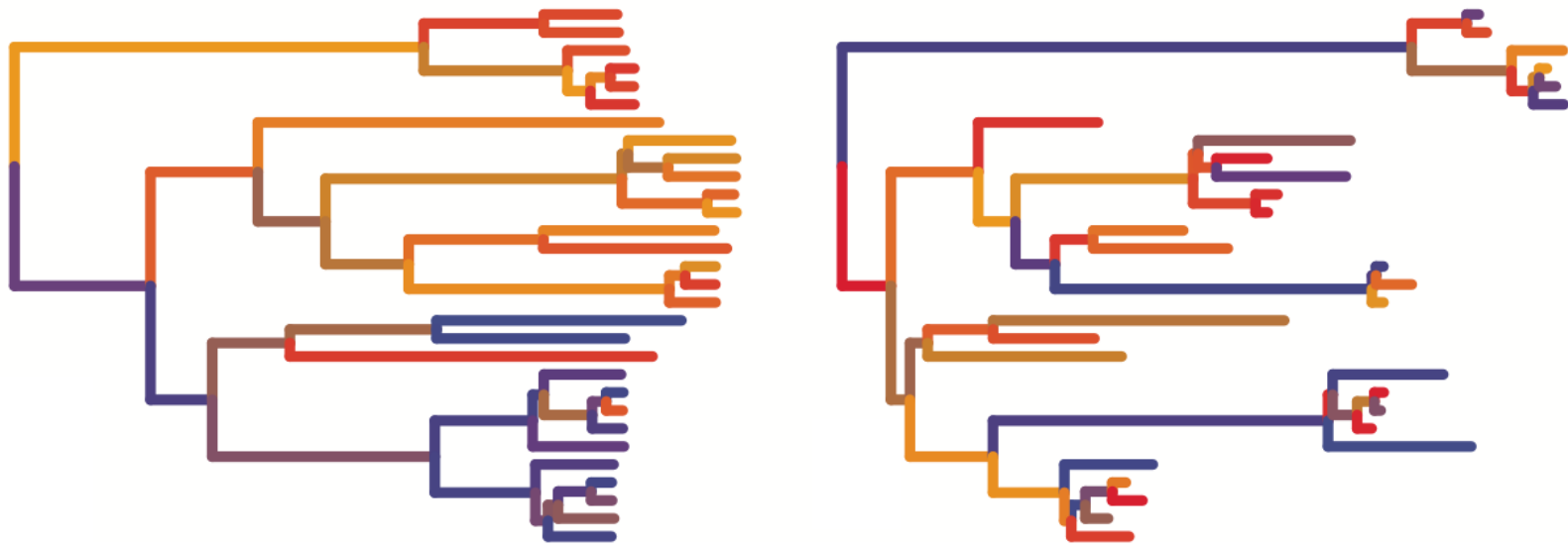
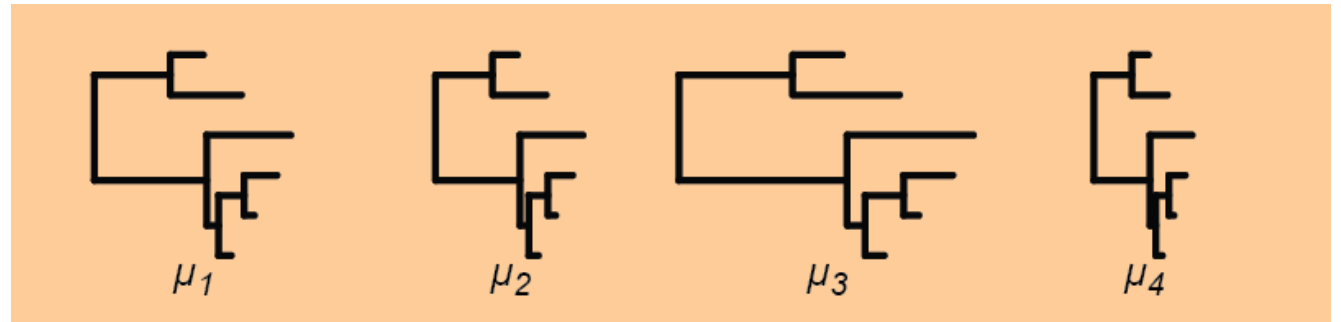
across loci



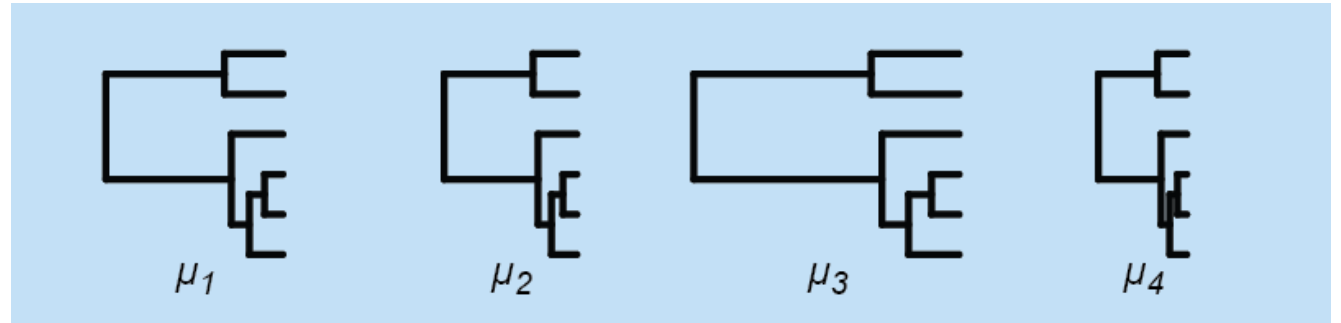
across loci



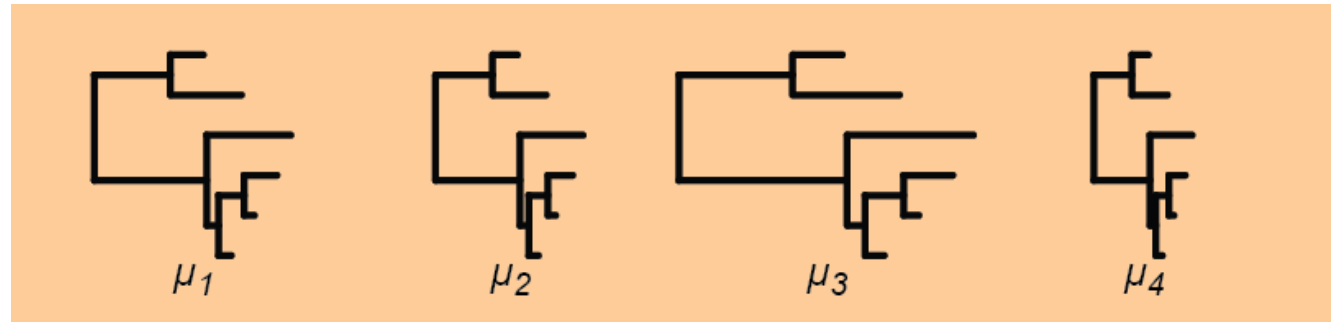
across lineages



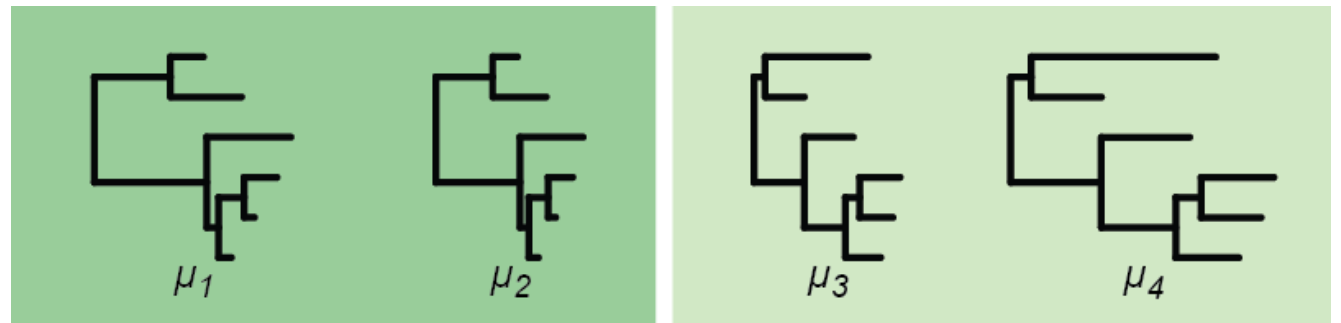
across loci



across lineages



gene-by-lineage
interactions

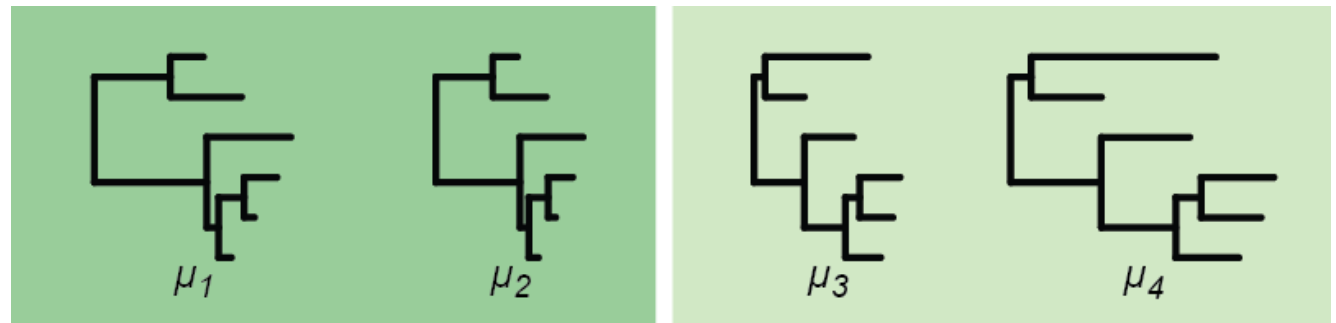


ClockstaR: choosing the number of relaxed-clock models in molecular phylogenetic analysis

Sebastián Duchêne^{*,†}, Martyna Molak and Simon Y. W. Ho[†]

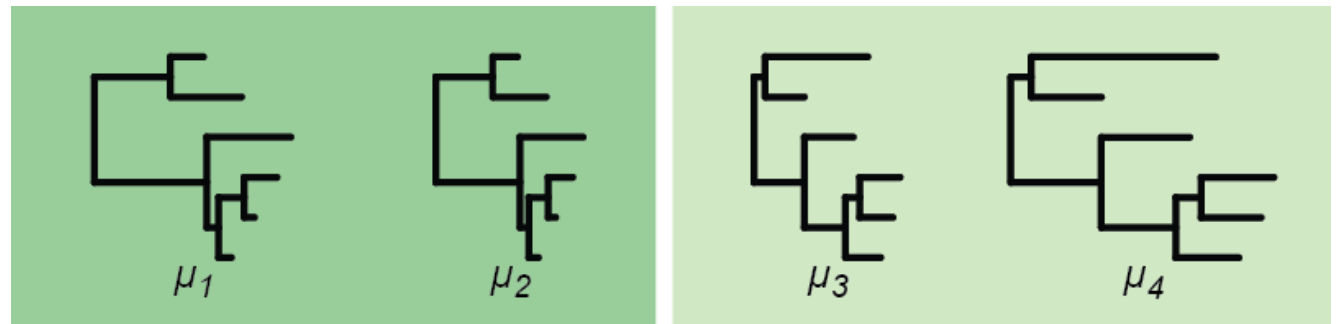
School of Biological Sciences, University of Sydney, Sydney, NSW 2006, Australia

gene-by-lineage
interactions



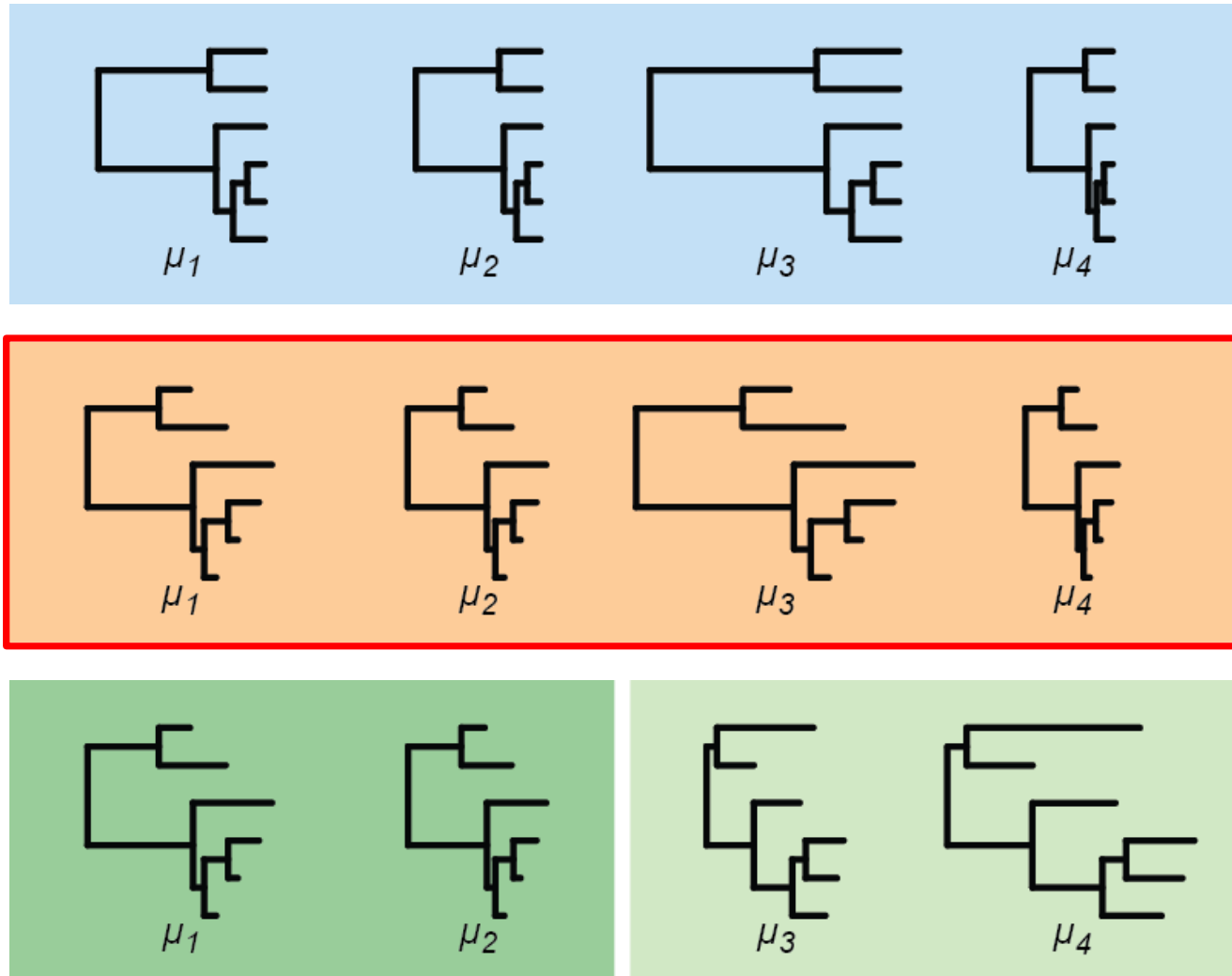
- 13 clock models for mammalian genomes
- 1–3 clock models for chloroplast genomes of flowering plants

gene-by-lineage
interactions



Partitioning the data for clock models makes little difference

Phylogenomic clocks

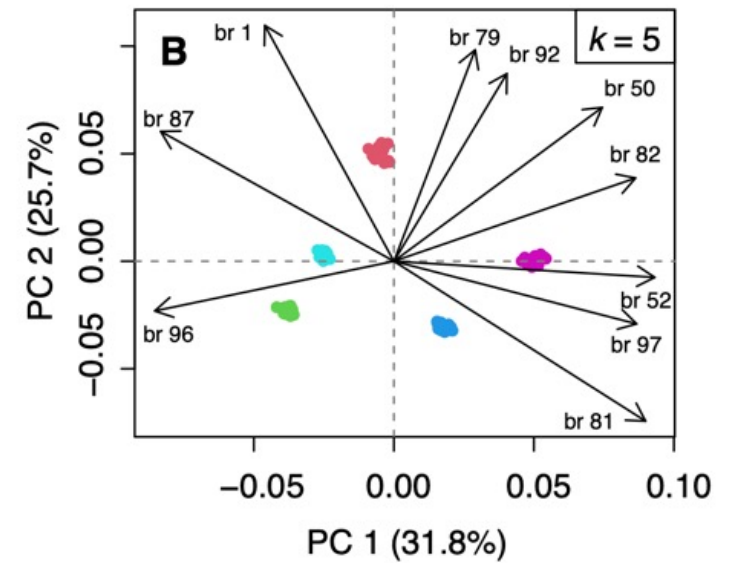
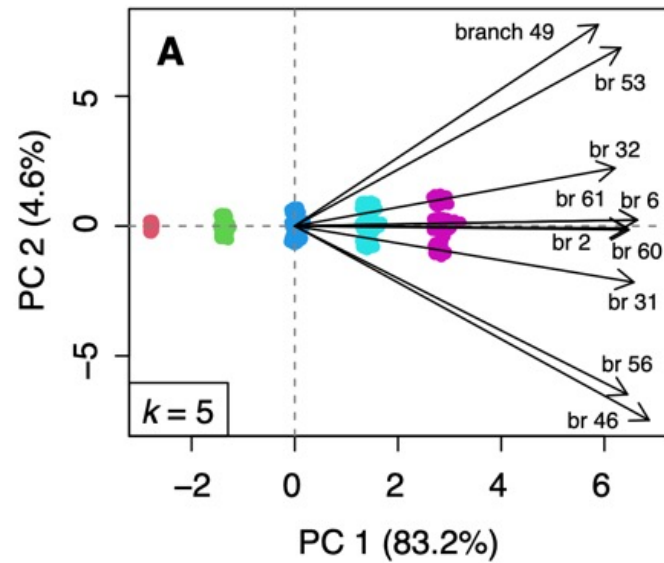


across loci

ClockstaRX: testing molecular clock hypotheses with genomic data

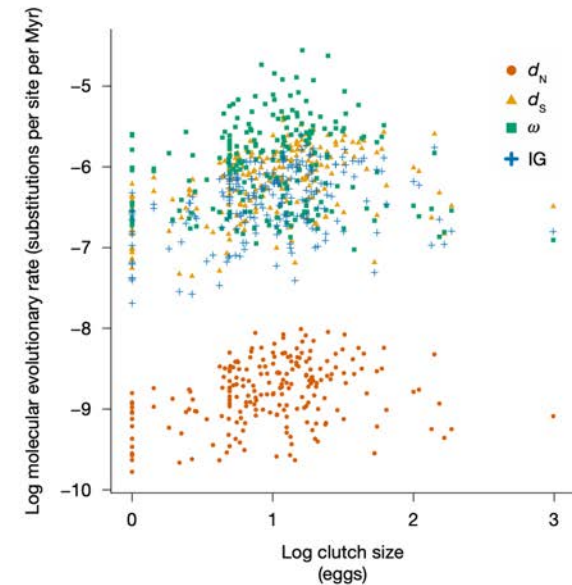
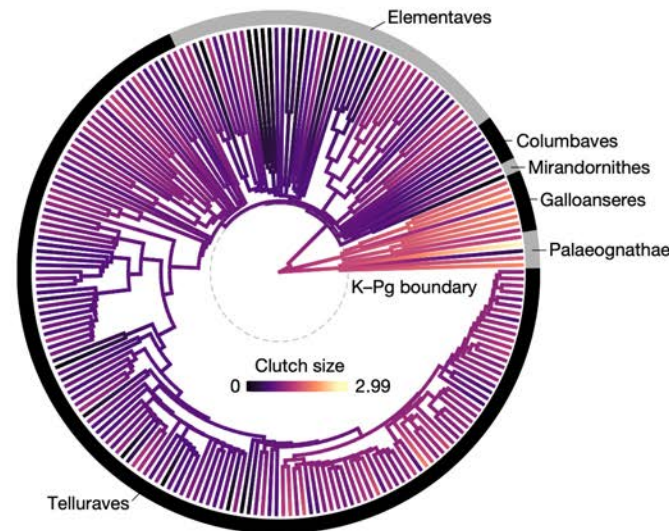
across lineages

gene-by-lineage
interactions

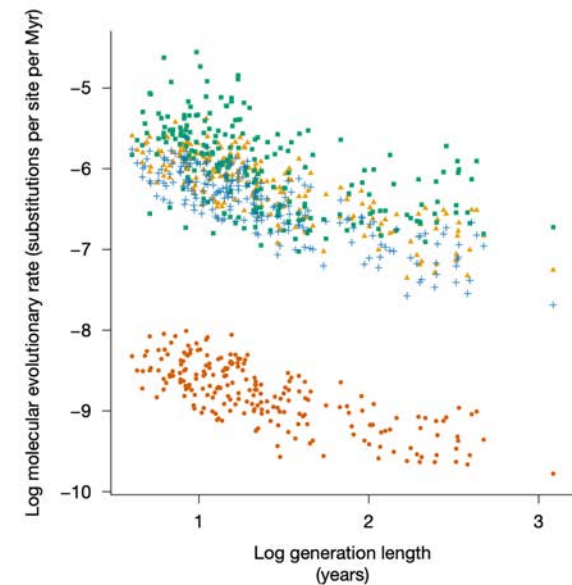
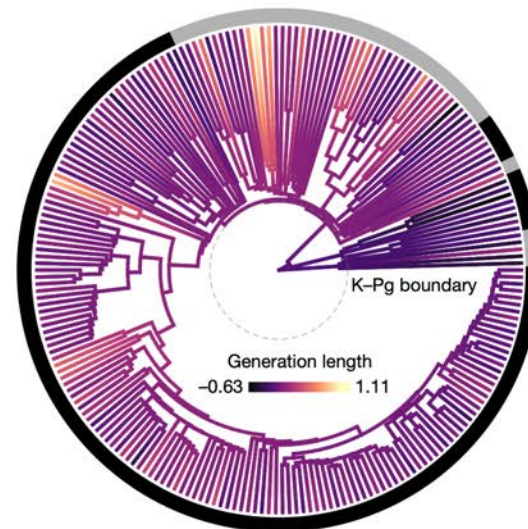


Evolutionary rates in birds

Clutch size



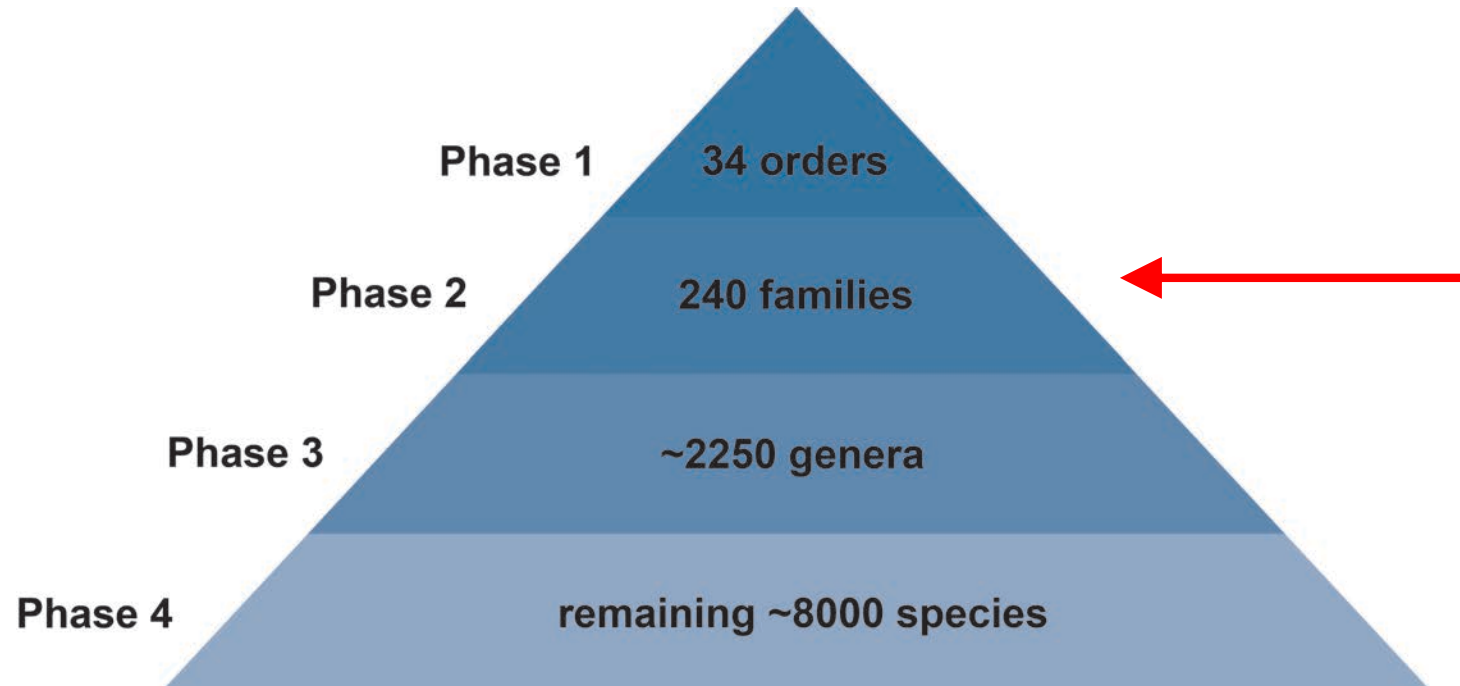
Generation length



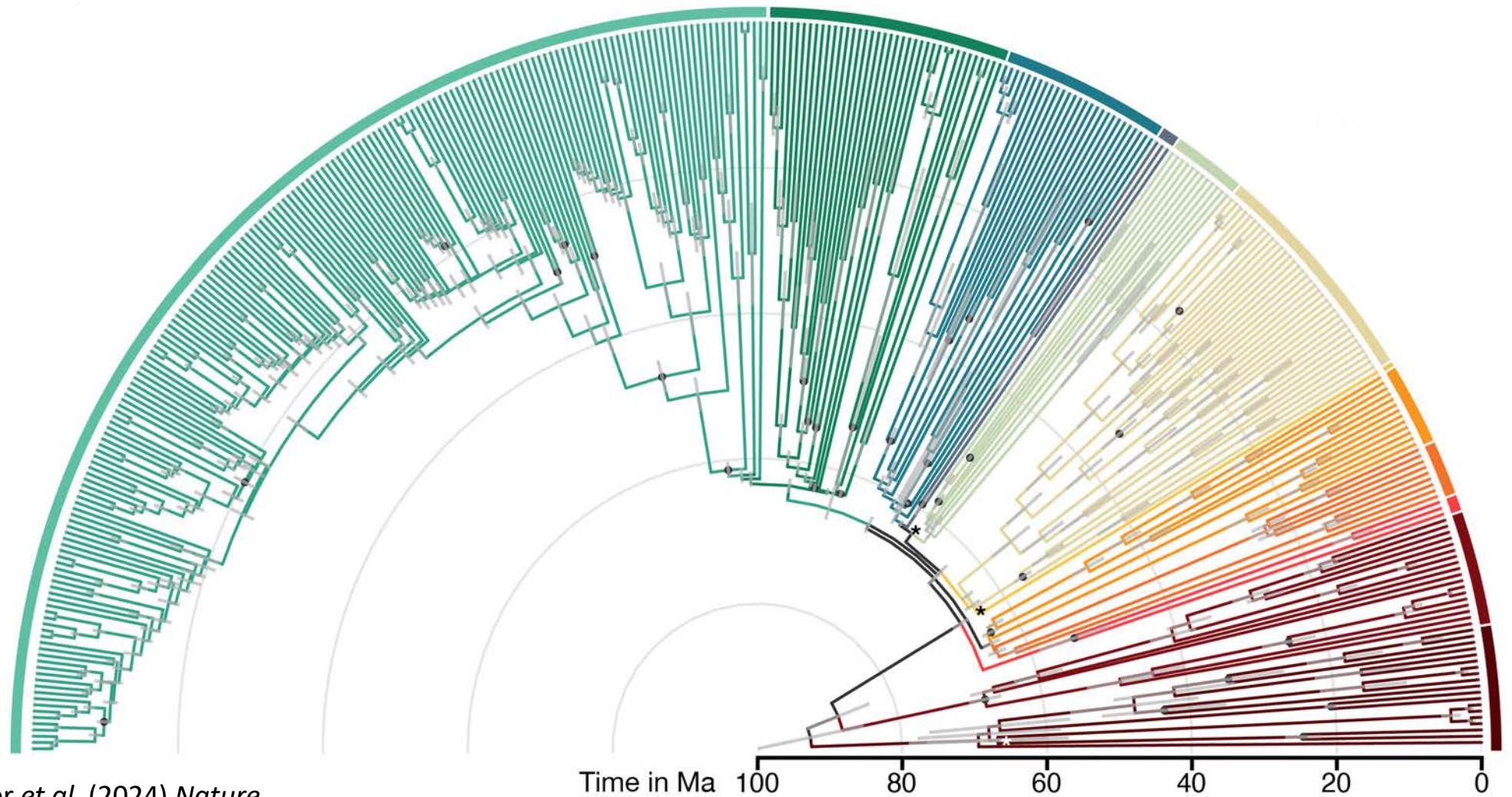
5. Example: Evolution of birds

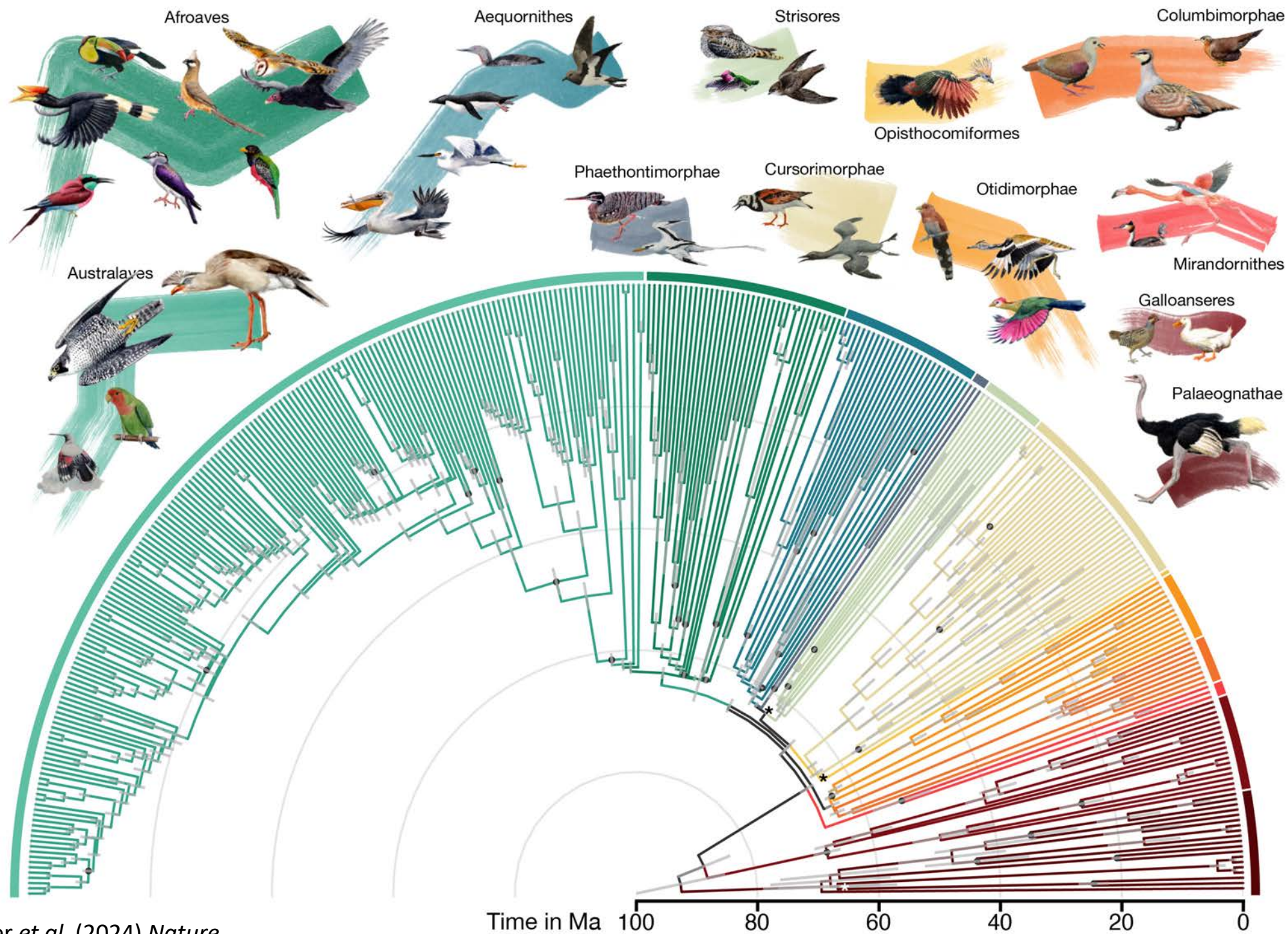
Phylogenomic study of 363 bird genomes



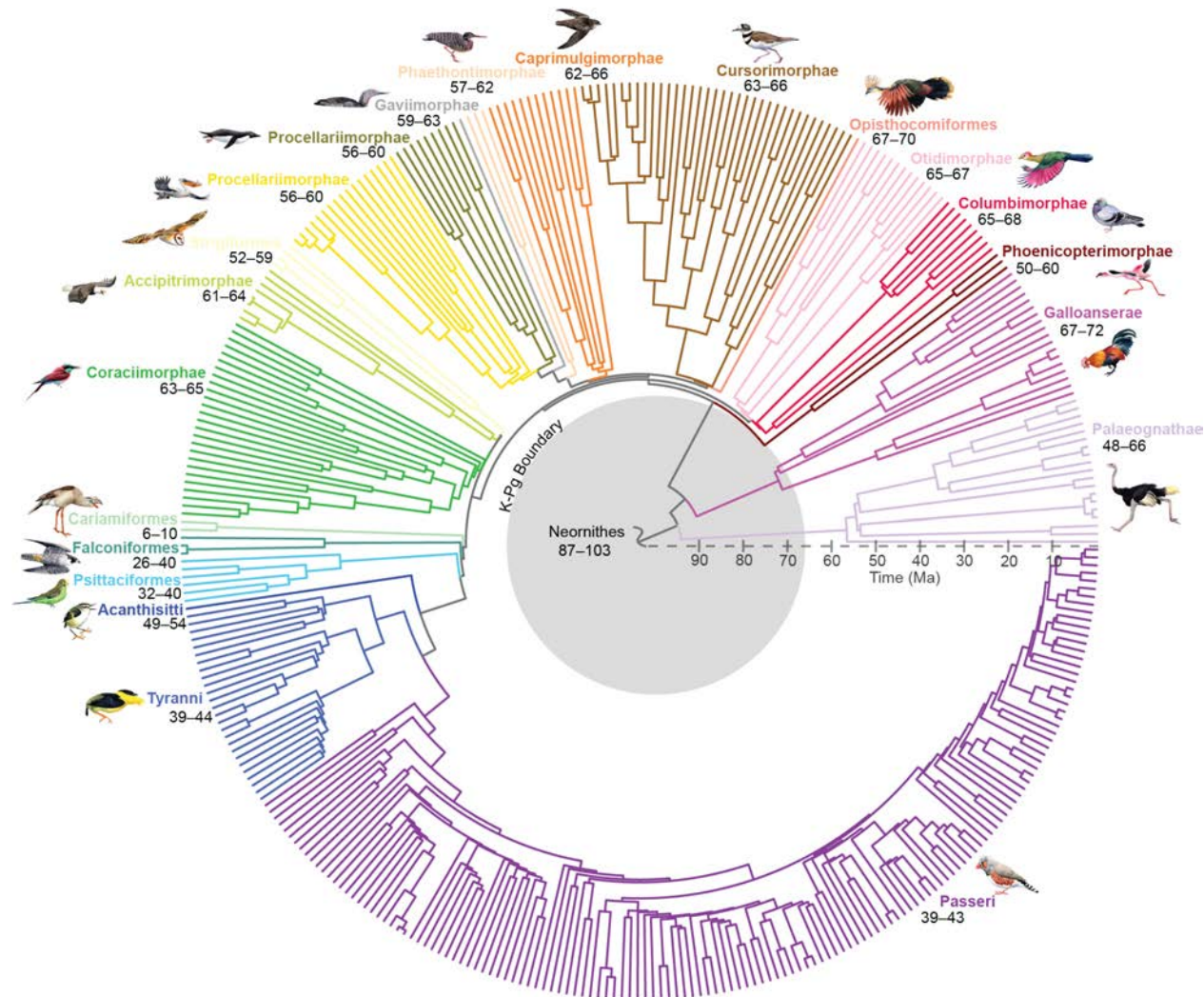


- 363 species
- 63,430 intergenic loci

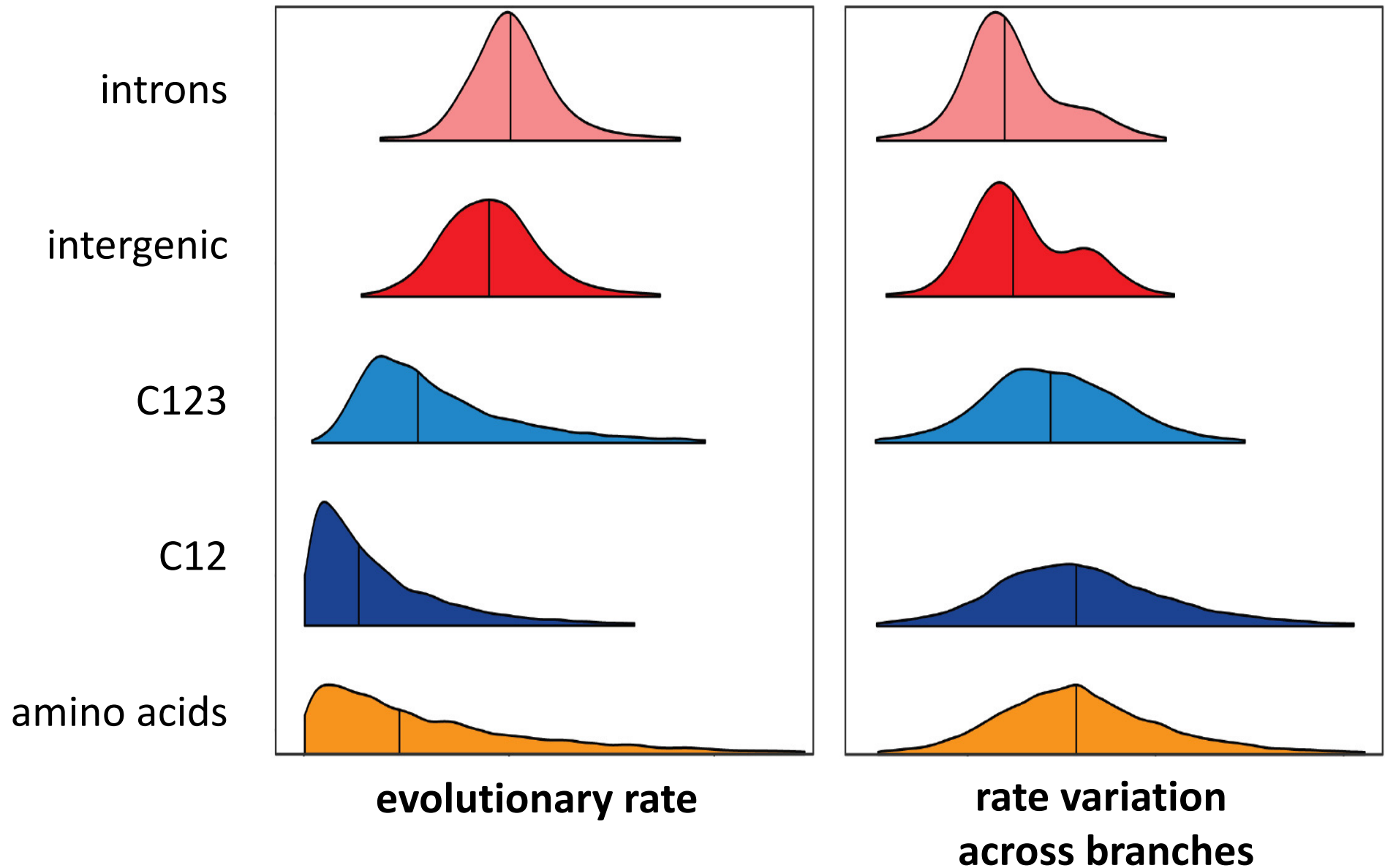




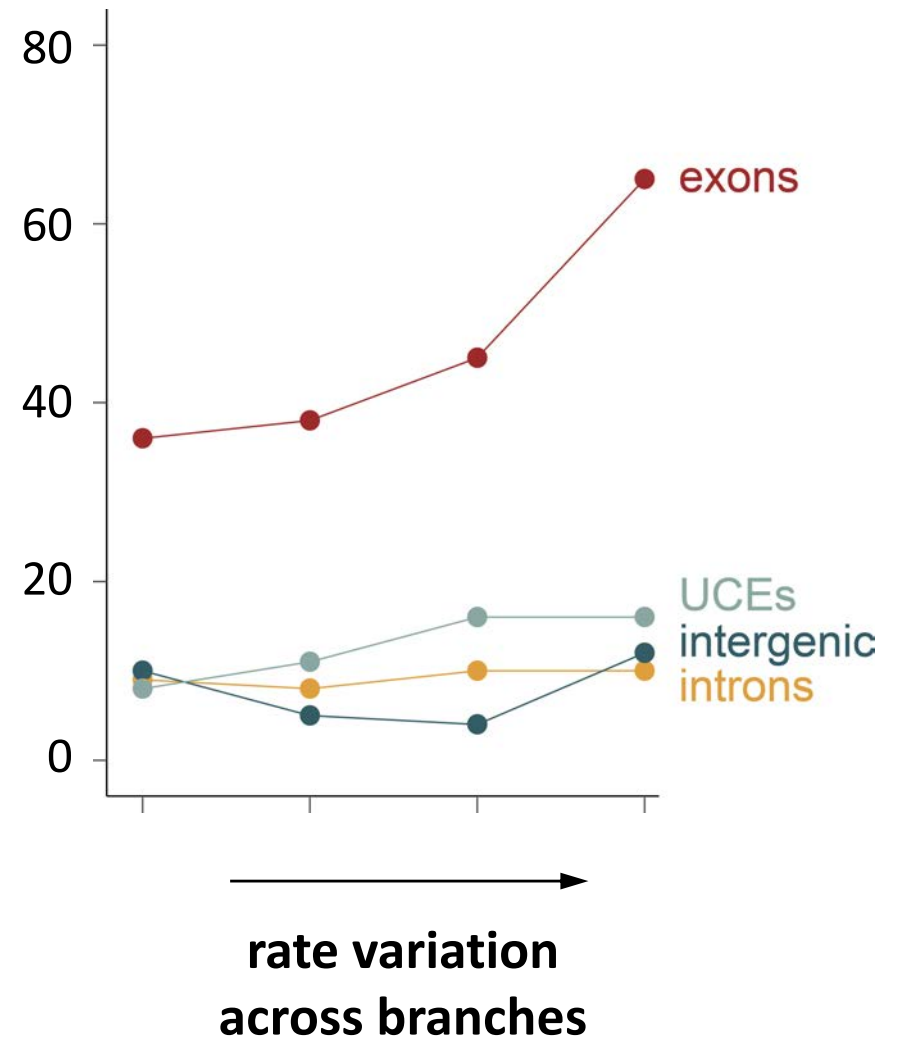
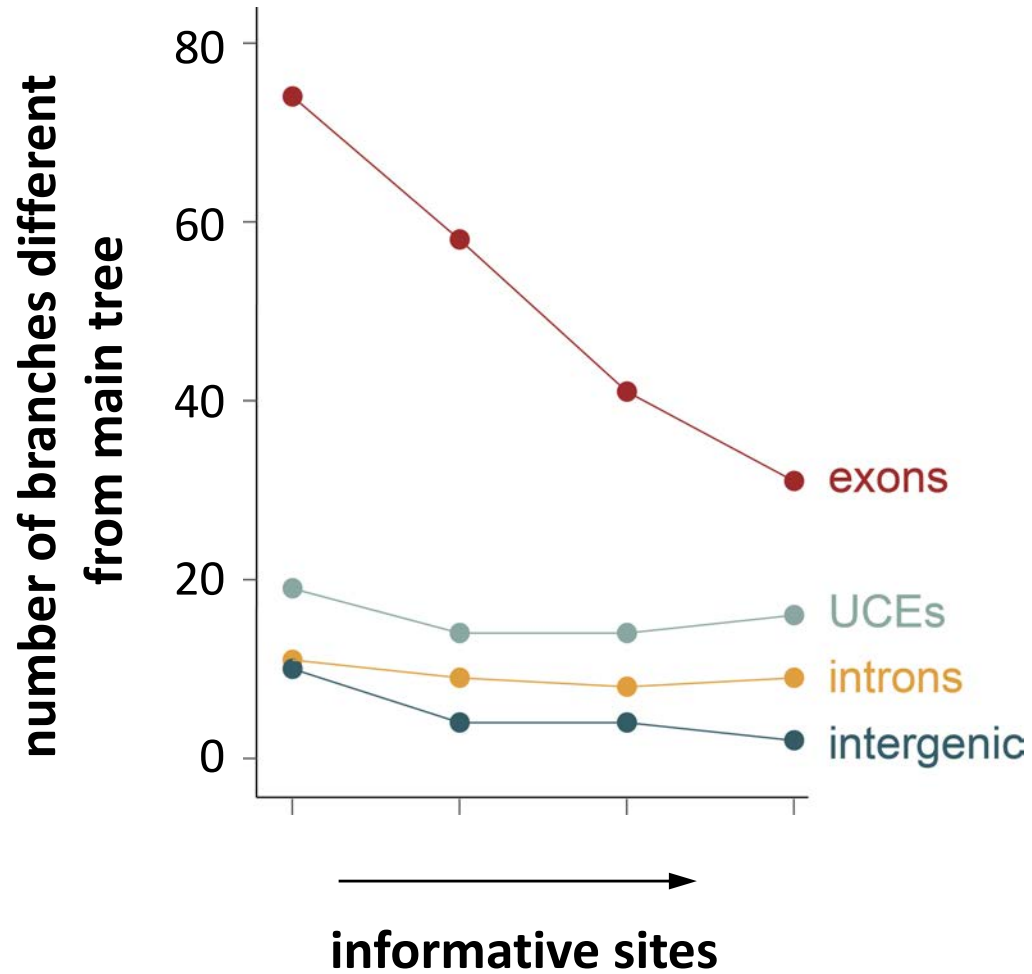
- Molecular dating using 10,494 loci
- Filtered by overall rate and degree of rate variation



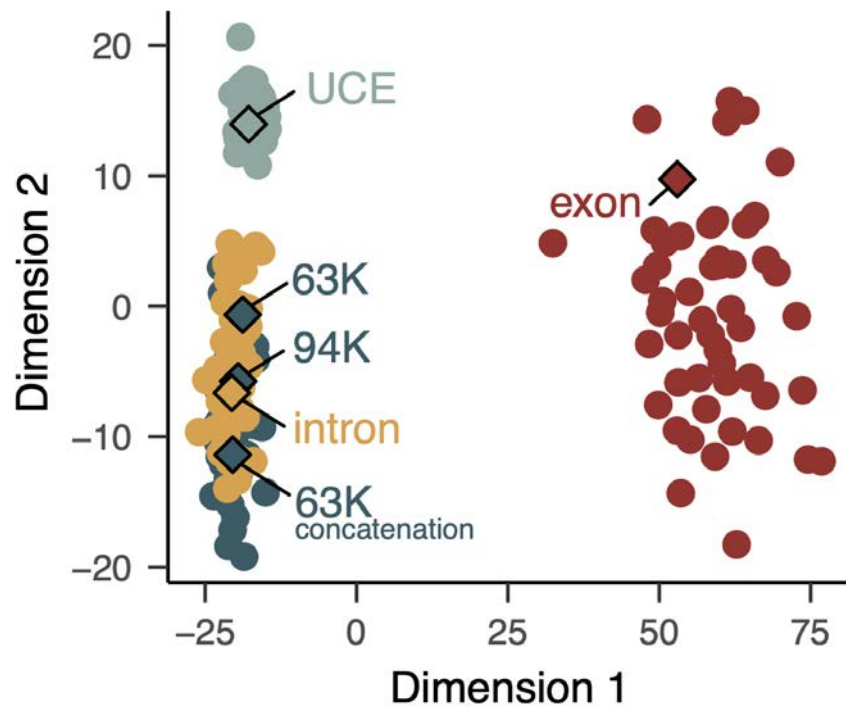
Genomic data types



Genomic data types



Genomic data types

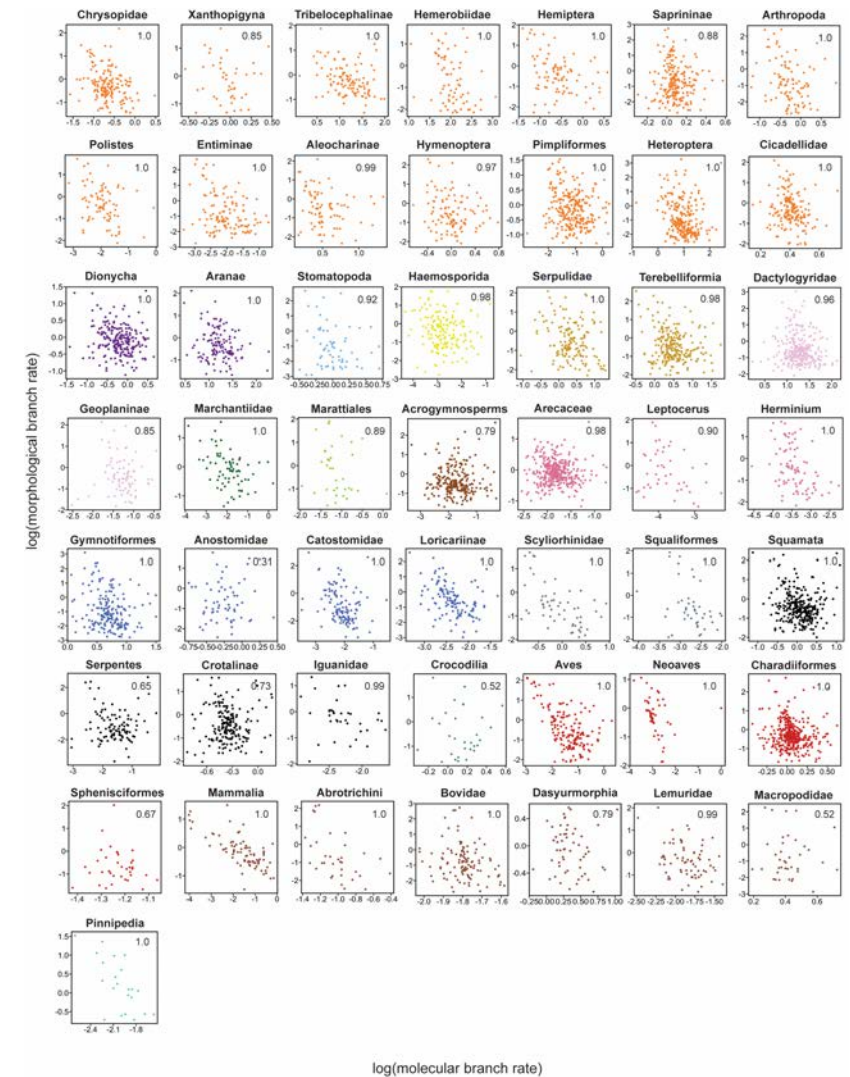


Problems with exons

- Lots of substitution saturation
- Compositional heterogeneity
- Rate variation across branches

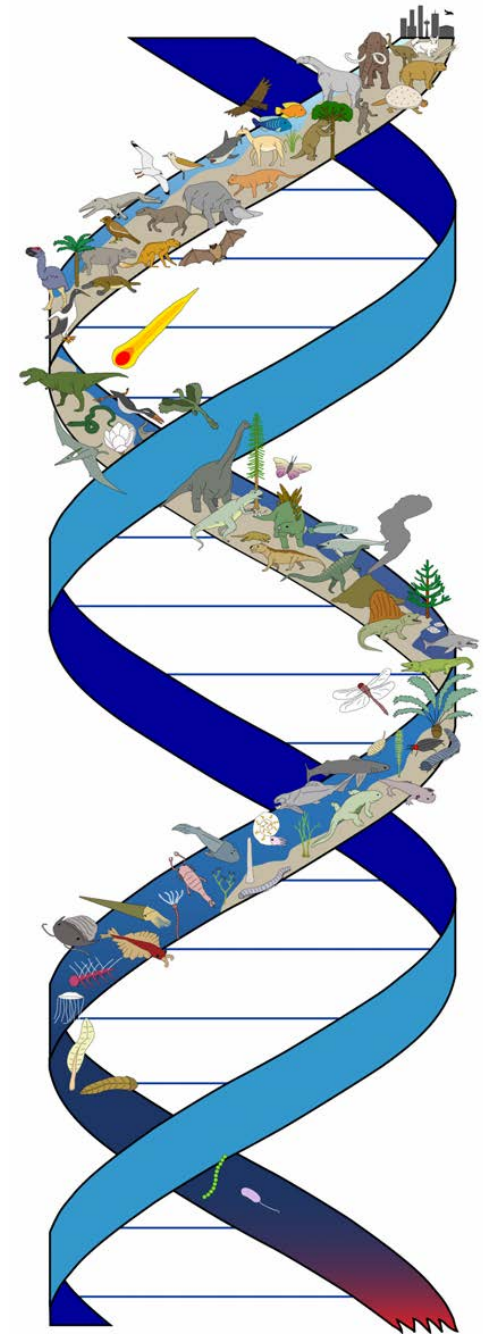
Phylogenomics and morphology

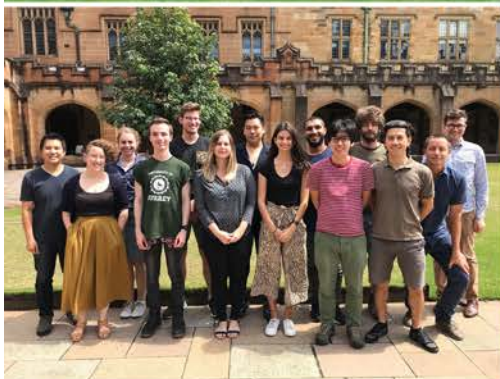
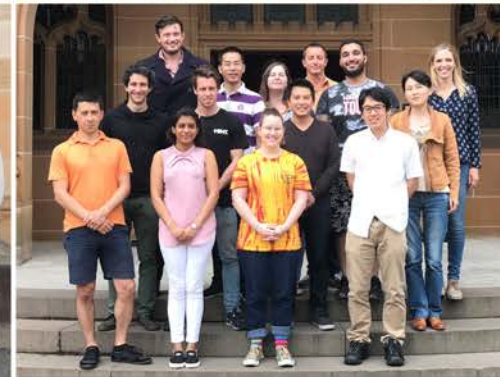
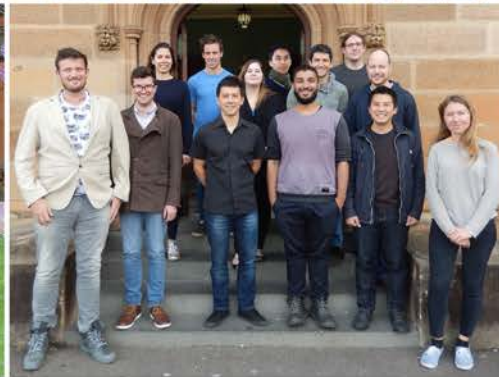
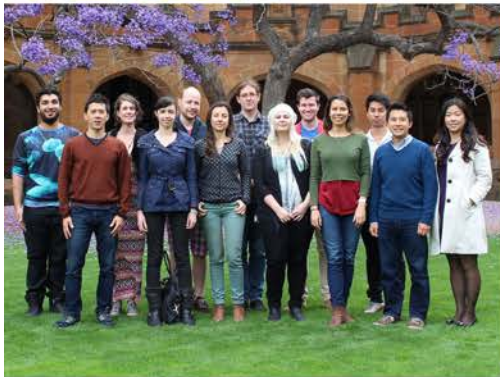
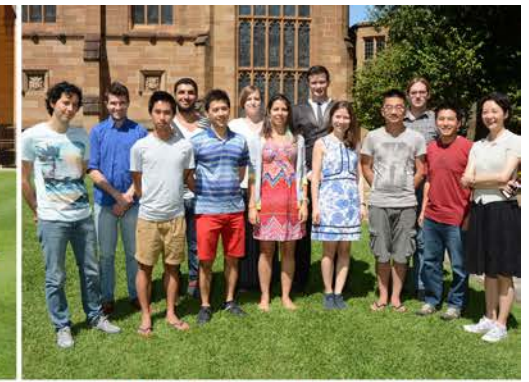
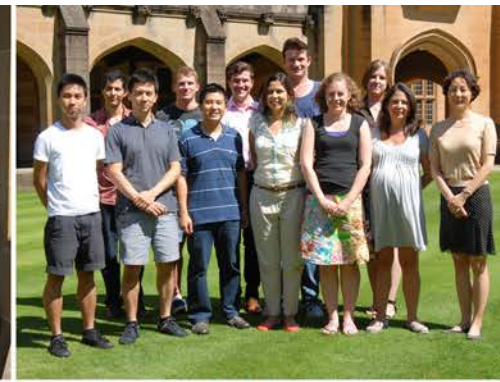
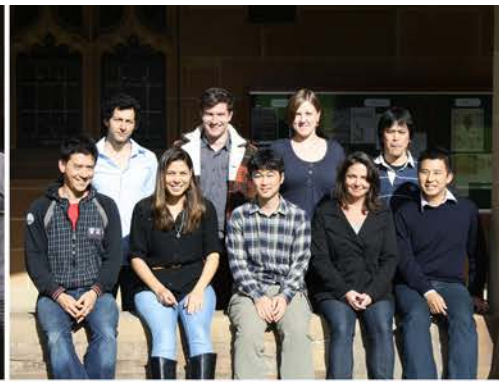
- Test for correlations between rates of morphological and molecular evolution
- Rates of genomic and morphological evolution are decoupled
- Morphological characters show much higher levels of rate variation across lineages



Key points

- Phylogenomics: Hundreds to thousands of genes
- Considerable computational demand
- Evolutionary complexity
- Resolution of evolutionary relationships and timescales
- Detailed insights into evolutionary processes





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